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Master Thesis

Title of the Thesis

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A short dedication placed on its own page, where I have no idea who
to dedicate this too...

Abstract

Will write the abstract here (about 300–500 words). Explain the problem, methodology, main contributions, and results.

Key words: keyword1, keyword2, keyword3

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CHAPTER 1

Introduction

Modern football data analysis increasingly depends on the ability to compare players and teams in a principled, scalable manner. Clubs, analysts, and recruitment departments are often faced with questions that are conceptually simple yet technically demanding: *Which players are truly similar in style? How does a player's role change across seasons or clubs? Can we characterise tactical profiles in a way that is both interpretable and statistically grounded?* Recent work in sports analytics and the broader literature on tactical behaviour highlights the growing importance of data-driven representations of playing style, collective dynamics, and contextual effects [24, 19]. However, many operational approaches remain either heuristic or too tightly tied to one-off visualisations, making it difficult to quantify similarity consistently across large populations, seasons, and evolving contexts.

This masters' thesis is motivated by a practical and scientific aim: to build a framework for *modelling* latent football style from player performance data using a hierarchical Bayesian model. Rather than attempting to predict a single outcome (e.g., goals or match results), the focus is on constructing a space in which players (and, ultimately, groups such as teams or positional archetypes) can be compared meaningfully. The Bayesian approach adopted here provides, as a direct consequence rather than an added feature, posterior uncertainty on the quantities of interest — the player/season/residual variance decomposition, the canonical factor loadings, and each player's latent position — rather than point estimates alone [23]. While Bayesian hierarchical modelling has a long track record for football outcome prediction [2] and performance modelling [20], comparatively less attention has been paid to Bayesian latent-space constructions tailored to similarity and style, particularly with hierarchical structure reflecting repeated measures across seasons.

The starting point of this work is a longitudinal dataset of player match statistics from LaLiga, covering twelve seasons (2011/12 to 2022/23) and aggregated to the player–season level. Each observation corresponds to a player in a given season, with a minimum exposure threshold to reduce noise from small samples. From these aggregates, we construct feature vectors designed to be comparable across players and seasons: volume and intensity measures expressed per 90 minutes, efficiency ratios (such as passing accuracy or duel win rates), and interpretable blocks reflecting common scouting and coaching perspectives (offensive production; progression and creation; carrying and 1v1 actions; defence and pressing; duels and physical contests). This block structure is not merely cosmetic, but rather facilitates encoding a domain-informed decomposition of performance into components that are often discussed qualitatively, and it provides a natural handle for weighting or extending the dataset when needed.

The resulting player–season observations are inherently multivariate, with substantial dependence structure across variables. Many football actions co-occur mechanically or tactically, and performance indicators often reflect overlapping dimensions of role, team context, and style. In such settings, analysing each variable in isolation can obscure the broader structure of the data, while naive comparisons in the original feature space may overstate noise, redundancy, or scale effects. A central methodological idea of this thesis is therefore to seek lower-dimensional latent representations that preserve the most relevant structure for comparison and interpretation. In practical terms, the goal is to represent player profiles in a way that is compact enough to visualise and reason about, yet rich enough to retain meaningful distinctions in style, usage, and tactical function.

From this perspective, the thesis is situated within *latent space modelling*. The main idea is that observed football statistics may be viewed as noisy, high-dimensional manifestations of a smaller number of underlying

dimensions related to playing style, role execution, and tactical behaviour. In this work, factor-analytic representations provide the route used to extract interpretable latent structure from correlated performance variables. This is especially useful in a longitudinal setting, where repeated player–season observations motivate extensions with hierarchical structure and raise substantive questions about persistence, evolution, and context dependence in football style.

The overarching objectives of this master’s thesis are therefore:

- (1) to construct a robust and interpretable representation of player performance profiles at the player–season level, grounded in football-relevant blocks of execution;
- (2) to study how multivariate football data can be represented in lower-dimensional latent spaces without losing their most important stylistic and tactical information;
- (3) to investigate factor-analytic latent-variable approaches for the representation and comparison of player profiles;
- (4) to provide Bayesian posterior uncertainty for the latent-space estimation, rank selection, and variance decomposition used throughout;
- (5) to frame the resulting representations as practical tools for scouting, player profiling, and tactical interpretation in modern football analytics [24, 19].

The remainder of this work is organised as follows. Chapter 2 describes the construction of the player–season dataset, from the raw match-event data through feature engineering to the final analytical sample. Chapter 3 introduces the exploratory methodology used to establish that the resulting feature space contains meaningful lower-dimensional structure, including classical factor analysis and multidimensional scaling. Chapter 4 specifies the hierarchical Bayesian factor model used for the main analysis: its additive decomposition, covariance structure, identification strategy, and rank selection. Chapter 5 reports model diagnostics, the resulting canonical factor directions and player archetypes, and case studies on player similarity and its scouting applications, and closes with the model’s limitations and directions for future work, connecting this line of work to broader Bayesian modelling in sports [2, 20, 23].

Part of the R and Stan code supporting this thesis — model implementation, debugging, and pipeline refactoring — was developed with the assistance of an AI coding assistant (Claude, Anthropic), used under the author’s direction and review throughout. The statistical modelling decisions, the analysis, and the interpretation of results are the author’s own.

CHAPTER 2

Data source and construction

The pillars of this master’s thesis is a longitudinal player–season dataset built from Wyscout football data. Wyscout data are event-based: match actions are recorded and tagged, and later summarised into match-level player statistics [27]. The data used here were accessed through a professional Redshift warehouse in a football industry setting, with access facilitated through Sebastien Coustou at Parma Calcio. The starting point was not a ready-made player-style table, but a staged match–player table that required filtering, aggregation, and feature engineering before any modelling itself.

The scope of the dataset is deliberately restricted to historical LaLiga data. This is not intended as a deficiency of the analysis, but as a design choice. The goal of the thesis is to study latent structure over time within a single competition, thereby avoiding the additional layer of heterogeneity introduced by cross-league differences in tempo, officiating, tactical distributions, data coverage, and competitive context. Extending the framework to several leagues is possible and potentially valuable, but such an extension would constitute a modelling problem of its own, rather than a harmless enlargement of the sample.

The final analytical object is a matrix of player–season observations. Each row corresponds to one outfield player in one LaLiga season, and each column corresponds to an engineered football feature. After filtering and aggregation, the working dataset contains $N = 4586$ player–season observations, $I = 1529$ unique players, and $P = 48$ transformed football features. Player identifiers, player names, season identifiers, competition identifiers, minutes played, and matches were retained as metadata, but were not used as active variables in the feature matrix for PCA, factor analysis, dissimilarity construction, or the Bayesian model of Chapter 4.

Operationally, the starting point was a staged Redshift table at the match–player level, with one row per player and match. This table contained event-derived counts, success counts, averages, and percentages. It was joined to auxiliary competition, season, and player metadata tables in order to identify LaLiga seasons and recover player-level descriptors. The variables available at this stage included goals, assists, shots, passes, progressive passes, dribbles, duels, aerial duels, recoveries, pressing duels, touches in the box, and expected-goals-related quantities. The extraction and aggregation logic is sketched in Appendix B.

The staged data covers twelve LaLiga seasons available in the extraction, from 2011/12 through 2022/23. Goalkeeper observations were removed before constructing the similarity representation. Rather than relying only on position labels, match-level rows with goalkeeper-specific activity (shots faced, saves, conceded goals, clean sheets, or goalkeeper exits) were excluded. This restriction is methodological: goalkeepers occupy a different action space from outfield players, so including them would make part of the distance structure reflect positional incomparability rather than stylistic similarity among comparable profiles.

The remaining match–player rows were aggregated to the player–season level. Match-level quantities were summed or combined into season totals for minutes, matches, starts, goals, assists, shots, expected goals, expected assists, passes (including forward, back, lateral, long, smart, progressive, and through passes), crosses, dribbles, progressive runs, accelerations, defensive actions (interceptions, clearances, blocked shots, sliding tackles), duels (defensive, offensive, aerial, pressing, and loose-ball), recoveries, opponent-half recoveries, possession losses, touches in the box, fouls, and cards. A minimum exposure filter of 600 minutes was then imposed. This reduces the influence of player–season observations whose per-90 or ratio summaries are unstable because they are based on very limited playing time.

From the aggregated player–season table, the final modelling features were constructed using five pre-processing steps.

DEFINITION 2.1 (Per-90 normalisation). For count variables, per-90 features were computed as

$$x_{ij}^{(90)} = 90 \frac{x_{ij}}{\text{minutes}_i}.$$

This places players with different playing time on a comparable scale and measures action intensity rather than season accumulation.

DEFINITION 2.2 (Efficiency ratios). Where a meaningful denominator was available, success or efficiency variables were computed as

$$r_{ij} = \frac{\text{successful actions}_{ij}}{\text{attempted actions}_{ij}}.$$

Examples include pass accuracy, progressive-pass accuracy, dribble success, defensive-action success, duel-win rate, aerial-win rate, and pressing-duel win rate. These variables separate involvement volume from execution quality.

DEFINITION 2.3 (Log transformation). For a small number of strongly right-skewed non-negative volume variables, a $\log(1 + x)$ transformation was applied:

$$x_{ij}^{(\log)} = \log(1 + x_{ij}).$$

This reduces the influence of extreme observations while preserving the ordering of non-negative values. Five features received this transform: goals, expected goals per shot, expected assists, shot assists, and key passes, all per 90 minutes.

DEFINITION 2.4 (Square-root transformation). A second group of moderately right-skewed volume variables received a milder $\sqrt{x}$ transformation instead:

$$x_{ij}^{(\sqrt{\cdot})} = \sqrt{x_{ij}}.$$

The square root compresses the upper tail less aggressively than $\log(1 + x)$ and was preferred for variables whose skew, while still non-negligible, did not warrant a logarithmic transform. Eighteen features received this transform, spanning most of the passing volume block (total passes, forward, back, lateral, and long passes, received passes, crosses, through passes, progressive passes, smart passes) together with shots, headed shots, shot blocks, recoveries, clearances, interceptions, and fouls suffered, all per 90 minutes. The remaining twenty-five features were left untransformed.

DEFINITION 2.5 (Standardisation). The (possibly transformed) features were standardised across the player–season sample using

$$z_{ij} = \frac{x_{ij} - \bar{x}_{\cdot j}}{s_j},$$

where $\bar{x}_{\cdot j}$ and s_j are the sample mean and standard deviation of feature j on the transformed scale. This prevents variables measured on larger numerical scales from dominating covariance- or distance-based analyses, and is applied after the log/square-root transforms above rather than to the raw counts.

For the exploratory descriptive analyses of Chapter 3, the engineered variables were additionally grouped into five football blocks: offensive production, progression and chance creation, carries and 1v1 actions, defence and pressing, and duels/physical contests (with the small number of discipline-related features folded into the duels/physical block). Equal block weighting was applied so that blocks with more variables did not dominate the representation purely because of dimensionality. Missing values in the final feature matrix were handled at the matrix-construction stage before computing correlations, dissimilarities, and latent representations.

TABLE 1. Summary of the final player–season analytical dataset.

Item	Description
Data provider	Wyscout-derived event data
Storage layer	Professional Redshift warehouse
Competition	LaLiga
Seasons	2011/12 – 2022/23 (12 seasons)
Observational unit	Player–season
Player population	Outfield players only
Minimum exposure	600 minutes per player–season
Number of observations	4586 player–seasons
Number of unique players	1529
Number of active features	48 engineered football features (5 log-transformed, 18 square-root-transformed, 25 untransformed, all Z-scaled)
Metadata retained	Player ID, player name, season ID, competition ID, minutes, matches
Active modelling matrix	Transformed, standardised football features (additionally block-weighted for the exploratory analyses of Chapter 3)

The output of the pipeline is a transformed player–season feature matrix rather than a raw event table. Each row corresponds to one outfield player in one LaLiga season, and each active column corresponds to an engineered football feature after normalisation, transformation, and standardisation. A summary of the final analytical dataset is given in [Table 1](#). This matrix is the input for the dissimilarity calculations and exploratory low-dimensional analyses of Chapter 3, and for the Bayesian hierarchical model of Chapters 4–5.

CHAPTER 3

Background

This chapter introduces the statistical background needed to study similarity between football players from a multivariate latent-space perspective. The main object of interest in this master's thesis is the *player–season observation*, that is, the aggregated statistical profile of a given player over a given season. Each observation is therefore described by multiple performance variables simultaneously, and the central challenge is to understand how such multivariate profiles can be represented, compared, and interpreted in a lower-dimensional space.

More broadly, the aim is to construct statistical representations of player performance that make stylistic similarity empirically assessable. In this setting, similarity is not understood as a directly observed quantity, but as a latent relational structure induced by the joint behaviour of many football variables. This naturally places the problem within the scope of multivariate analysis, where the goal is often to reduce complexity while preserving the main patterns of variation, dependence, and association across observations.

The player–season construction also introduces a repeated-observation component, since some players appear in more than one season. This suggests that player profiles may exhibit persistence or variation over time. This feature of the data is kept in mind as a possible motivation for richer latent-variable formulations if the descriptive and probabilistic analyses indicate that such extensions are warranted.

From a modelling perspective, the chapter develops the background in three parts. First, we review the general multivariate setting of the problem and the idea of representing high-dimensional football data in a smaller number of latent dimensions. Second, we discuss classical low-dimensional descriptive tools—in particular principal component analysis, factor analysis, and classical multidimensional scaling—which are useful for exploratory structure discovery and for identifying interpretable geometric or latent patterns in the data. Third, we introduce probabilistic counterparts to these descriptive ideas, with particular attention to Bayesian latent-space formulations that can incorporate uncertainty directly into the representation.

The need for this background is already visible in the exploratory analysis conducted on the player–season dataset. The data is built from event-based football statistics provided by *Wyscout*, a professional football data platform that collects and structures detailed match event information for scouting and performance analysis. From these event records, season-level aggregates are constructed and organised into the football blocks we defined previously. Once such multivariate profiles are defined, a number of statistical questions arise naturally: how many latent dimensions are needed to summarise the main structure, whether the variables reflect a smaller number of underlying stylistic components, and whether the resulting representation is stable and interpretable enough to support meaningful comparisons between players.

Beyond their descriptive value, these representations may also be relevant for downstream applications in recruitment and squad planning. A low-dimensional latent representation of player–season profiles can provide a structured way to compare footballers, identify stylistic neighbours, and understand how players relate to broader positional or tactical archetypes. In this thesis, several descriptive approaches (principal component analysis, factor analysis, and multidimensional scaling) are explored as candidate tools for this purpose, after which greater emphasis is placed on probabilistic analogues that can represent latent structure while also quantifying uncertainty.

1. Similarity representation

In multivariate analysis, comparing observations requires some rule for translating several variables into a single notion of proximity. This can be done through methods that work directly with covariance or correlation structure, or through a similarity or dissimilarity measure defined on the observed feature vectors. In either case, the underlying issue is the same: different comparison rules emphasize different aspects of the data and may therefore induce different low-dimensional representations.

This point is especially relevant when observations are interpreted as performance profiles. Two player–season observations may be regarded as similar because they have similar magnitudes across variables, because they exhibit similar relative patterns across variables, or because they align along a common latent direction even if their absolute levels differ. Thus, similarity is not a uniquely defined property of the data, but rather a statistical construct that depends on how the multivariate profile is represented and compared.

In the present work, the player–season observations are built from weighted football features summarising different aspects of performance. After suitable preprocessing, standardisation, and block-wise weighting, two comparison rules are considered as initial descriptive choices. The first is the Euclidean distance between observations $x_i, x_j \in \mathbb{R}^p$,

$$d_E(x_i, x_j) = \left(\sum_{k=1}^p (x_{ik} - x_{jk})^2 \right)^{1/2}.$$

This measure is sensitive to overall discrepancies in the feature space and therefore reflects differences in absolute profile location after the chosen transformations have been applied.

The second is a cosine-based dissimilarity, defined as

$$d_C(x_i, x_j) = 1 - \frac{x_i^\top x_j}{\|x_i\| \|x_j\|}.$$

This quantity depends on the angle between the two feature vectors and therefore emphasises profile orientation relative to the origin rather than absolute separation in feature space. Strictly speaking, d_C is not a metric distance: $d_C(x_i, x_j) = 0$ implies that x_i and x_j have the same direction, but not necessarily that $x_i = x_j$.

This distinction is important because this cosine dissimilarity and Pearson correlation use different reference points. Cosine dissimilarity compares vectors relative to the origin, whereas correlation compares centred profiles. In particular, if $\tilde{x}_i = x_i - \bar{x}_i$ and $\tilde{x}_j = x_j - \bar{x}_j$, then

$$\text{corr}(x_i, x_j) = \frac{\tilde{x}_i^\top \tilde{x}_j}{\|\tilde{x}_i\| \|\tilde{x}_j\|}.$$

Thus, correlation is the cosine similarity between centred profiles. This makes correlation relevant for the later PCA and factor-analytic modelling, since these methods are directly based on covariance/correlation structure.

These two measures are not intended to exhaust the set of reasonable comparisons in multivariate statistics. Rather, they provide two conceptually distinct and practically useful benchmarks for the exploratory stage of the analysis: one based more directly on geometric separation in the transformed feature space, and one based on angular separation between multivariate profiles. Their comparison is informative precisely because the resulting similarity structures need not coincide.

The exploratory results in [Figure 1](#) show that Euclidean distance and cosine dissimilarity are substantially related, but not equivalent. This is expected because both are computed from the same transformed feature vectors, while emphasising different aspects of them. Euclidean distance reflects absolute separation in the weighted standardised feature space, whereas cosine dissimilarity reflects angular separation relative to the

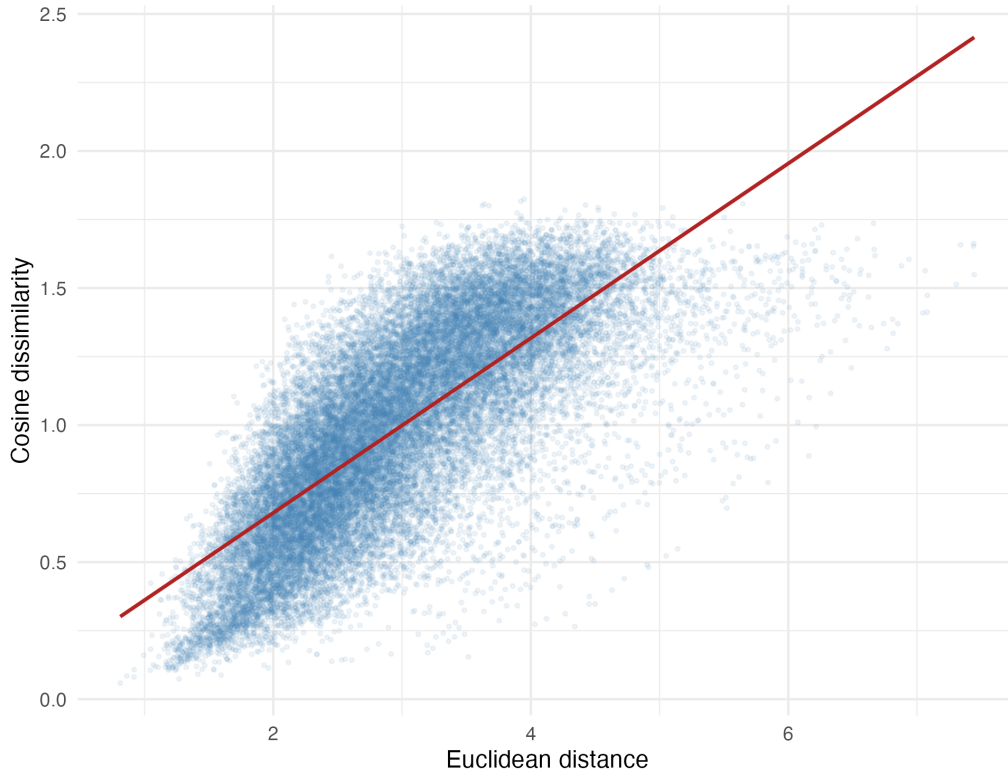


FIGURE 1. Pairwise Euclidean vs. cosine dissimilarities for the player-season subset ($N = 900$, 30,000 sampled pairs). The two metrics show moderate-to-strong agreement (Pearson = 0.746, Spearman = 0.774).

origin. Consequently, the induced neighbourhood structure and the interpretation of player similarity may change depending on the comparison rule.

2. Exploratory low-dimensional structure

A natural question in multivariate analysis is whether a high-dimensional set of observations can be represented satisfactorily in a smaller number of dimensions without losing its main structural patterns. In the present work, this question is approached first through classical exploratory techniques, namely principal component analysis (PCA), factor analysis (FA), and classical multidimensional scaling (CMDS) [14, 3, 4]. These methods do not provide the final probabilistic framework of the work, but they are useful diagnostic tools for determining whether the feature space and the induced distance structure contain stable lower-dimensional patterns worth modelling.

Before specifying a richer latent-variable model, it is helpful to understand whether the data exhibits interpretable low-dimensional structure at all, whether such structure is approximately stable across relevant subsets of it, and whether the dominant structure is more naturally described through covariance/correlation patterns or through dissimilarities. In particular, exploratory analyses can be stratified or inspected by season, playing time, or positional group in order to assess whether a single global representation is plausible or whether the data suggest more heterogeneous structure that should later guide the modelling strategy.

These methods serve different but complementary purposes. PCA works directly on the feature matrix and identifies orthogonal linear combinations that explain as much sample variation as possible [14]. FA also works on the feature matrix, but seeks a lower-rank latent structure capable of reproducing the dependence

among the observed variables up to variable-specific residual variation [3]. CMDs, by contrast, starts from a matrix of pairwise dissimilarities and searches for an embedding whose interpoint distances approximate the observed dissimilarities [4]. Taken together, these methods help address several preliminary questions: whether dimension reduction is plausible, whether the dominant axes or factors admit football interpretation, whether the conclusions are robust across relevant subsets of the data, and whether the structure suggested by the features is compatible with the structure suggested by the dissimilarities.

2.1. Principal component analysis. Let $\mathbf{X} = (x_{ij}) \in \mathbb{R}^{n \times p}$ denote the feature matrix, where n is the number of player–season observations and p is the number of transformed and standardised football features retained for the analysis. In the present application, $n = 4586$, corresponding to player–season rows, and the columns of $\mathbf{X}$ represent season-level statistics derived from Wyscout event data, including variables related to shooting, finishing, progression, passing, dribbling, defensive actions, recoveries, duels, and aerial involvement.

Principal component analysis (PCA) is a classical dimension-reduction technique that seeks linear combinations of the observed variables with maximal sample variance under orthogonality constraints [14]. It is useful here as an exploratory device because it reveals whether the player–season feature space is dominated by a small number of broad directions of variation, and because it helps identify which football variables contribute most strongly to those directions.

To define PCA formally, let $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$ denote the rows of $\mathbf{X}$, and assume that the columns of $\mathbf{X}$ have been centred. For any vector $\mathbf{v} \in \mathbb{R}^p$, define the projected sample scores

$$z_i(\mathbf{v}) = \mathbf{x}_i^\top \mathbf{v}, \quad i = 1, \dots, n,$$

and their sample variance

$$\widehat{\text{Var}}(z(\mathbf{v})) = \frac{1}{n-1} \sum_{i=1}^n (z_i(\mathbf{v}) - \bar{z}(\mathbf{v}))^2.$$

DEFINITION 3.1. The first principal component direction is defined as

$$\mathbf{v}_1 = \arg \max_{\|\mathbf{v}\|=1} \widehat{\text{Var}}(z(\mathbf{v})),$$

and, recursively, for $k \geq 2$, the k -th principal component direction is defined as

$$\mathbf{v}_k = \arg \max_{\substack{\|\mathbf{v}\|=1 \\ \mathbf{v} \perp \mathbf{v}_1, \dots, \mathbf{v}_{k-1}}} \widehat{\text{Var}}(z(\mathbf{v})).$$

The vectors $\mathbf{v}_1, \dots, \mathbf{v}_p$ are called the principal component directions, and the projected variables $\mathbf{X}\mathbf{v}_1, \dots, \mathbf{X}\mathbf{v}_p$ are called the principal component scores.

A standard result states that these directions are the eigenvectors of the sample covariance matrix of the centred data [14]. If

$$\mathbf{S} = \frac{1}{n-1} \mathbf{X}^\top \mathbf{X},$$

then the principal component directions are given by the eigenvectors of $\mathbf{S}$, ordered by decreasing eigenvalue. The corresponding eigenvalues $\lambda_1 \geq \dots \geq \lambda_p$ are the sample variances explained by the successive components. If the analysis is instead performed on standardised variables, then one may equivalently work with the sample correlation matrix

$$\mathbf{\Omega} = \frac{1}{n-1} \mathbf{Z}^\top \mathbf{Z}, \quad \mathbf{Z} = \left[\frac{\mathbf{X}_{\cdot 1} - \bar{X}_1 \mathbf{1}}{s_1}, \dots, \frac{\mathbf{X}_{\cdot p} - \bar{X}_p \mathbf{1}}{s_p} \right],$$

where $\bar{X}_j$ is the sample mean and s_j is the sample standard deviation of the j -th variable [14]. The coefficients of each eigenvector are commonly referred to as *loadings*, while the coordinates of the observations after projection onto the component directions are the *scores*.

PCA is used as an exploratory instrument applied only to the transformed football-feature matrix. Season identifiers, competition identifiers, player identifiers, player names, minutes played, and matches are not treated as active PCA variables; they are retained as metadata for interpretation and diagnostic checks. First, PCA provides a rough sense of whether the player–season feature space is intrinsically high-dimensional or whether a moderate number of directions captures a substantial proportion of sample variation. Second, the leading loadings indicate which football actions most strongly define the dominant axes of variation. Third, the component scores can be inspected across metadata-defined subsets—for instance by season, competition, or minutes played—to assess whether the same low-dimensional structure is broadly visible across the sample.

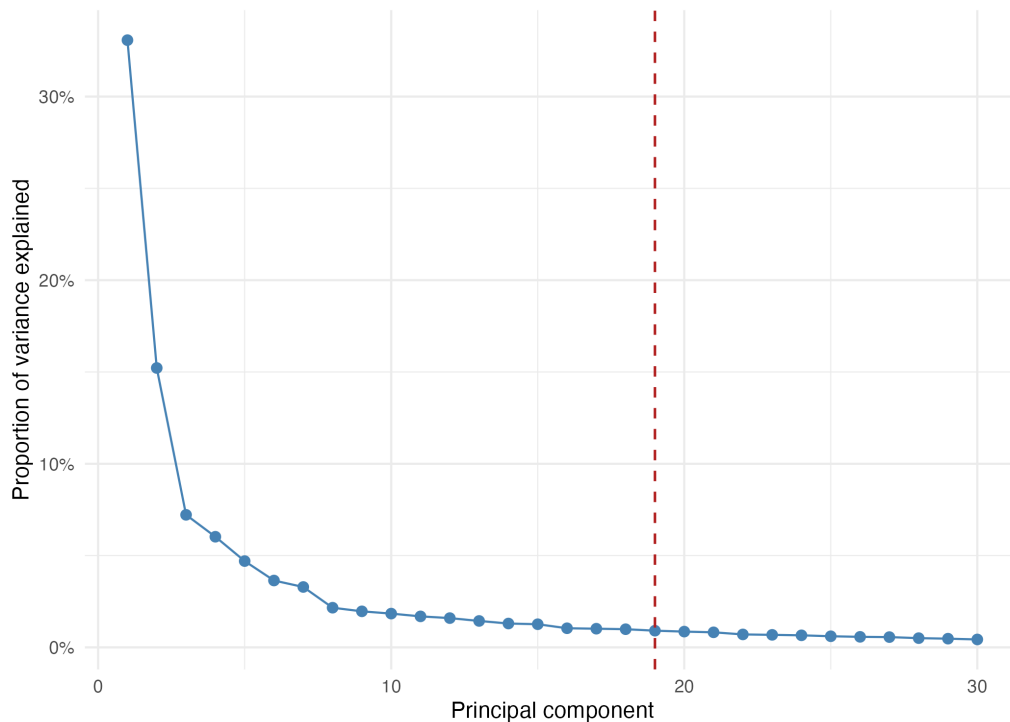


FIGURE 2. Scree plot of the first 30 principal components. PC1 and PC2 jointly explain approximately 48% of the variance, while the first ten components explain approximately 79%. The dashed line marks the first component count reaching at least 90% cumulative explained variance (nineteen components).

The exploratory results suggest that the data are structured but not trivially so. In particular, [Figure 2](#) shows that the first two principal components explain 48.3% of the total sample variance, the first five around 66.2%, and the first ten roughly 79.1%. Using 90% cumulative explained variance as a descriptive threshold requires about nineteen principal components for a high-fidelity linear summary. This indicates that a low-dimensional summary is plausible, but also that the player–season feature space cannot be reduced to a single simple axis without substantial information loss: with $P = 48$ engineered features spanning finishing, passing, carrying, defending, and duelling, considerably more shared structure remains beyond the first few directions than a smaller, coarser feature set would suggest. To interpret the leading components beyond their explained variance, the variables with the largest absolute loadings were inspected. [Table 1](#) reports the five largest absolute loadings for each of the first five principal components.

Interpretations based on the loadings are purely descriptive, and the signs of PCA loadings should not be over-interpreted because eigenvectors are identifiable only up to sign.

TABLE 1. Largest absolute PCA loadings for the first five principal components.

Component	Feature	Loading
PC1	per90_dribbles	0.318
PC1	per90_touch_in_box	0.251
PC1	per90_shots	0.244
PC1	per90_accelerations	0.229
PC1	per90_progressive_run	0.225
PC2	per90_head_shots	-0.298
PC2	per90_progressive_run	0.295
PC2	per90_accelerations	0.279
PC2	per90_aerial_duels	-0.242
PC2	per90_dribbles	0.215
PC3	per90_defensive_duels	-0.342
PC3	per90_fouls	-0.341
PC3	rate_successful_dribbles	0.303
PC3	per90_loose_ball_duels	-0.268
PC3	per90_dribbles_against	-0.263
PC4	rate_successful_dribbles	0.308
PC4	per90_accelerations	-0.299
PC4	per90_progressive_run	-0.286
PC4	per90_passes_to_final_third	0.228
PC4	per90_through_passes	0.223
PC5	rate_successful_dribbles	0.754
PC5	rate_offensive_duels_won	0.281
PC5	per90_accelerations	0.217
PC5	per90_progressive_run	0.177
PC5	per90_dribbles	0.171

Although PCA is informative, its limitations are important. Each component is constructed to be uncorrelated with the previous ones because PCA imposes orthogonality on the extracted directions. This is a mathematical constraint of the method, not evidence that the underlying football traits are independent. For example, ball progression, chance creation, and attacking involvement may be substantively related even if their PCA summaries are forced to lie on orthogonal axes. PCA is therefore interpreted here as a descriptive compression of the transformed feature space, not as a structural model of latent football style.

2.2. Factor analysis. Factor analysis provides an alternative route to low-dimensional structure when the observed variables are thought to reflect a smaller number of underlying latent traits. In contrast to PCA, which is defined through variance-maximising linear projections, factor analysis begins from an explicit statistical model for the covariance structure of the observed variables [3].

DEFINITION 3.2. A factor analysis model for observation i , represented by a random vector $\mathbf{Y}_i \in \mathbb{R}^p$, with k latent factors is given by

$$\mathbf{Y}_i = \boldsymbol{\mu} + \boldsymbol{\Lambda} \mathbf{f}_i + \boldsymbol{\varepsilon}_i,$$

where $\boldsymbol{\mu} \in \mathbb{R}^p$ is a mean vector, $\mathbf{f}_i \in \mathbb{R}^k$ is the latent factor vector for observation i , $\boldsymbol{\Lambda} \in \mathbb{R}^{p \times k}$ is a loading matrix, and $\boldsymbol{\varepsilon}_i \in \mathbb{R}^p$ is an observation-specific residual vector. The matrix $\boldsymbol{\Lambda}$ describes how the

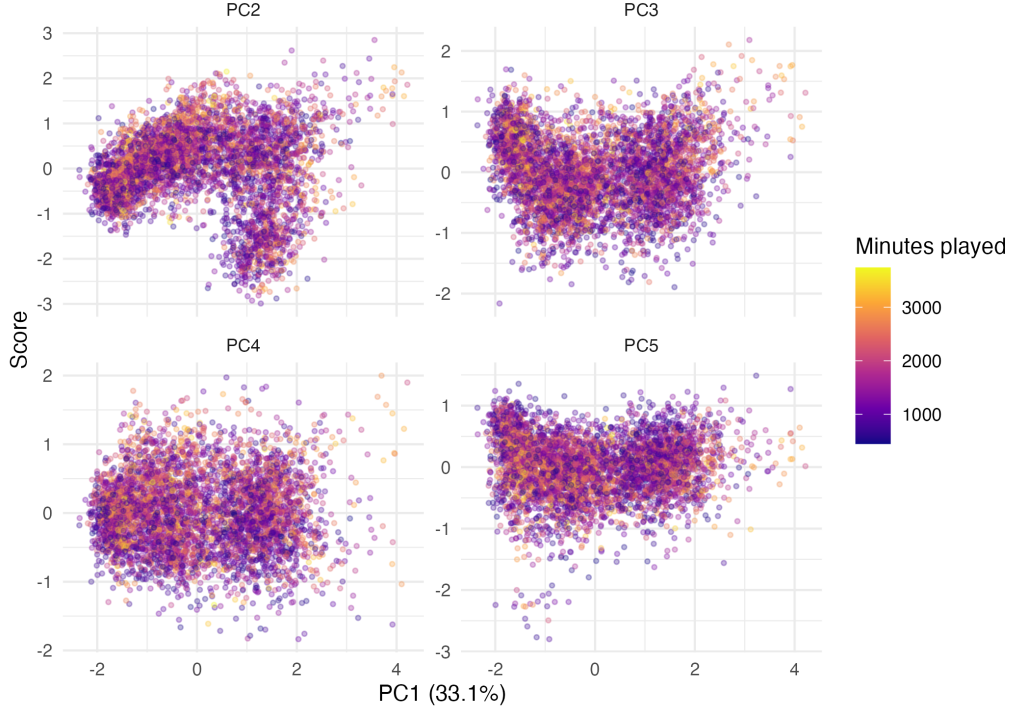


FIGURE 3. Principal component score plots for PC1 against PC2–PC5, with points coloured by minutes played. Playing time does not show a clear monotone gradient along the displayed components, suggesting that minutes played are not the main source of variation in the leading PCA scores.

observed variables depend linearly on the latent factors, while ε captures variable-specific residual variation not explained by the common factor structure [3].

Under standard assumptions

$$\mathbb{E}[\mathbf{f}_i] = \mathbf{0}, \quad \text{Cov}(\mathbf{f}_i) = \mathbf{I}_k, \quad \mathbb{E}[\varepsilon_i] = \mathbf{0}, \quad \text{Cov}(\varepsilon_i) = \Psi,$$

where $\Psi = \text{diag}(\psi_1, \dots, \psi_p)$, and $\mathbf{f}_i$ is independent of ε_i , it follows that

$$\text{Cov}(\mathbf{Y}) = \mathbf{\Lambda}\mathbf{\Lambda}^\top + \Psi.$$

Thus, the covariance matrix is decomposed into a low-rank component, $\mathbf{\Lambda}\mathbf{\Lambda}^\top$, representing shared dependence among the variables, and a diagonal component, Ψ , representing variable-specific residual variance.

This framework is appealing in football because the observed features are unlikely to be driven by wholly separate mechanisms. Variables related to progression and creation may share common structure, as may defensive activity and duel intensity, while finishing output may combine contributions from shot volume, box presence, and shot quality. In such a setting, a model that explains dependence through a smaller number of related latent traits is often more plausible than one based only on orthogonal descriptive directions.

Unlike PCA and CMDs, factor analysis is already a probabilistic model. For this thesis, factor analysis provides a simple latent-variable baseline for assessing whether the covariance structure of the player–season feature matrix can be represented through a number of interpretable latent dimensions before specifying a final Bayesian model.

In the exploratory analysis conducted here, after removing a constant feature with zero sample standard deviation, maximum-likelihood exploratory factor analysis was fitted for several candidate numbers of factors, followed by oblique rotation using an oblimin-type criterion in order to allow correlated factors [3, 13].

The number of factors was guided in part by parallel analysis [12]. Parallel analysis compares the observed eigenvalues of the empirical correlation matrix with eigenvalues obtained from random datasets of the same size. Factors are retained while the observed eigenvalues exceed a chosen simulated reference, here the simulated 95th percentile. As shown in Figure 4, the observed eigenvalues dominate the simulated reference eigenvalues up to the eleventh factor, after which the empirical curve falls below the simulated 95th percentile. This provides empirical support for retaining eleven factors as a reasonable working dimension for the latent covariance structure; the larger count relative to a coarser feature set is expected, since $P = 48$ engineered features resolve considerably more distinct football sub-skills (e.g. separate passing-distance, duel-type, and defensive-action categories) than a smaller feature set would. Competing specifications were then compared using standard fit diagnostics.

More specifically, the root mean square residual (RMSR) summarises the average size of the residual correlations left unexplained by the fitted factor model, while the root mean square error of approximation (RMSEA) measures lack of fit per degree of freedom at the covariance-structure level [3, 5]. The Tucker–Lewis index (TLI), also known as the non-normed fit index, compares the fitted model’s discrepancy per degree of freedom against that of a baseline model in which all variables are uncorrelated, so that values closer to 1 indicate a comparatively better, more parsimony-adjusted fit [26]. Model comparison also considered the Bayesian information criterion (BIC), defined in the sense of Schwarz [21], which balances model fit against complexity.

The exploratory results support the presence of non-trivial latent structure. Parallel analysis suggested an eleven-factor solution, while a nine-factor model provided a more compressed but rougher summary. The eleven-factor specification improved the fit statistics relative to the nine-factor one (TLI rising from 0.804 to 0.843; RMSEA falling from 0.101 to 0.090; RMSR falling from 0.0256 to 0.0227; BIC falling from approximately 28,630 to 19,420), although the overall fit still indicates that the latent structure should be viewed as an approximation rather than a complete explanation of the dependence in the data. Taken together, the pattern in Figure 4 and the fit comparisons suggest that eleven factors offer a reasonable compromise between parsimony and interpretability for this larger, $P = 48$ feature set.

Substantively, the rotated loading structure reveals several interpretable latent dimensions; see Figure 5. Ordered by the variance each factor accounts for, the eleven-factor solution separates: chance creation and assists; goal threat and finishing; carrying and offensive duelling; general passing volume and circulation; defensive duel-winning and ball recovery; defensive duel exposure (1v1 defending workload); progressive and vertical passing; physical engagement and possession risk; high/pressing recoveries; passing accuracy versus crossing risk; and deep build-up circulation. While these labels are necessarily interpretive, the loading pattern shows that the latent dimensions are not arbitrary mathematical artefacts, but correspond to recognisable football behaviours – including a clear separation, absent from a coarser feature set, between winning defensive duels and simply being exposed to them, and between general passing volume and specifically progressive, forward-directed passing. Importantly, the estimated factor correlations are not negligible (the strongest, between the chance-creation and carrying/offensive-duelling factors, reaches 0.63, and goal threat correlates negatively with both defensive factors at around -0.5), which reinforces the idea that football style may be better described through related latent traits than through a small set of orthogonal descriptive axes.

The role of factor analysis in this master’s thesis is therefore both exploratory and strategic. It helps assess whether the transformed feature matrix supports an interpretable latent covariance structure, and it provides guidance for the construction of more refined probabilistic models later in the work. At the same time, it remains distinct from approaches based directly on pairwise dissimilarities, so its conclusions should

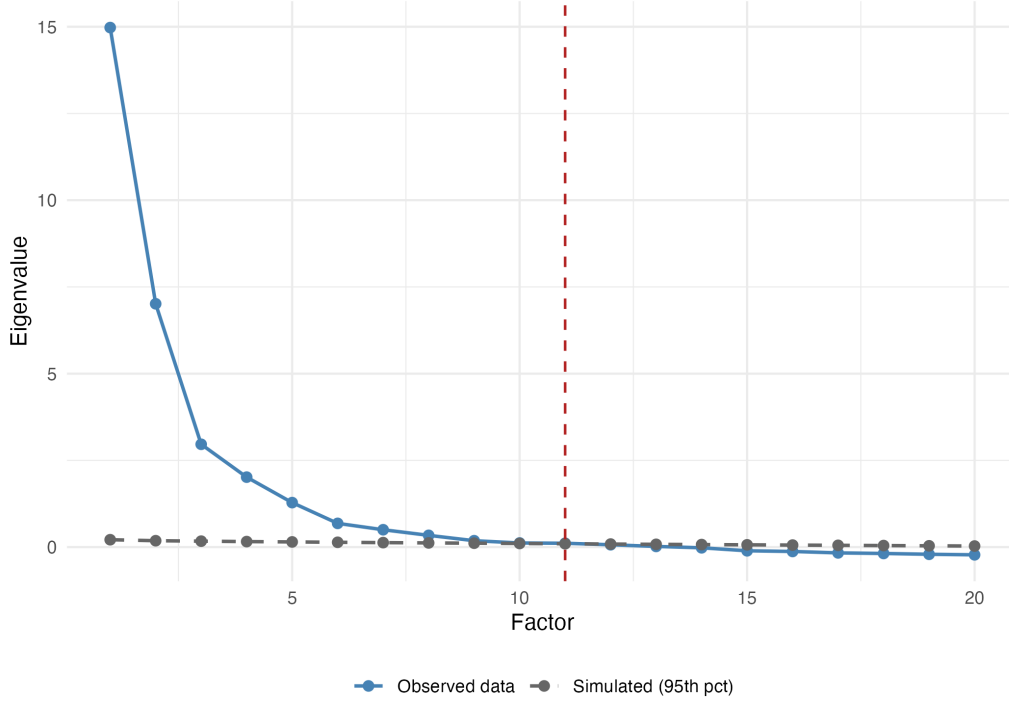


FIGURE 4. Parallel analysis scree plot for factor analysis. The observed eigenvalues are compared with the 95th percentile of eigenvalues obtained from simulated random datasets. The vertical dashed line indicates the suggested number of retained factors, $k = 11$, corresponding to the point after which the observed eigenvalues no longer exceed the simulated reference values.

be interpreted as one stage in a sequential modelling strategy rather than as the unique final representation of football similarity.

2.3. Classical multidimensional scaling. Classical multidimensional scaling (CMDS) starts from a matrix of pairwise dissimilarities rather than directly from the feature matrix. When those dissimilarities are ordinary Euclidean distances computed from centred feature vectors, CMDS is closely connected to PCA: the double-centred squared-distance matrix recovers the same principal-coordinate geometry as the PCA score representation. Its added value here is therefore not to duplicate PCA with Euclidean distances, but to examine non-Euclidean or profile-based dissimilarities, such as cosine-based dissimilarity. This makes it relevant here because one of the exploratory goals is to understand whether the similarity structure induced by the player–season representation can be expressed as a low-dimensional Euclidean configuration [25, 4, 7].

Let $\mathbf{D} = (d_{ij})$ denote the matrix of pairwise dissimilarities between the n observations, with $d_{ii} = 0$, $d_{ij} = d_{ji}$, and let $q < n$ be a target embedding dimension. The CMDS problem is to construct points $\mathbf{z}_1, \dots, \mathbf{z}_n \in \mathbb{R}^q$ such that the Euclidean interpoint distances $\|\mathbf{z}_i - \mathbf{z}_j\|$ approximate the observed dissimilarities as closely as possible. In its classical form, this is achieved through a spectral decomposition of a double-centred matrix of squared dissimilarities [25, 4].

THEOREM 3.3. *Suppose that $\mathbf{D}^{(2)} = (d_{ij}^2)$ is the matrix of squared pairwise Euclidean distances generated by some centred configuration of n points in $\mathbb{R}^m$. Let*

$$\mathbf{J} = \mathbf{I}_n - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$$

3. BACKGROUND

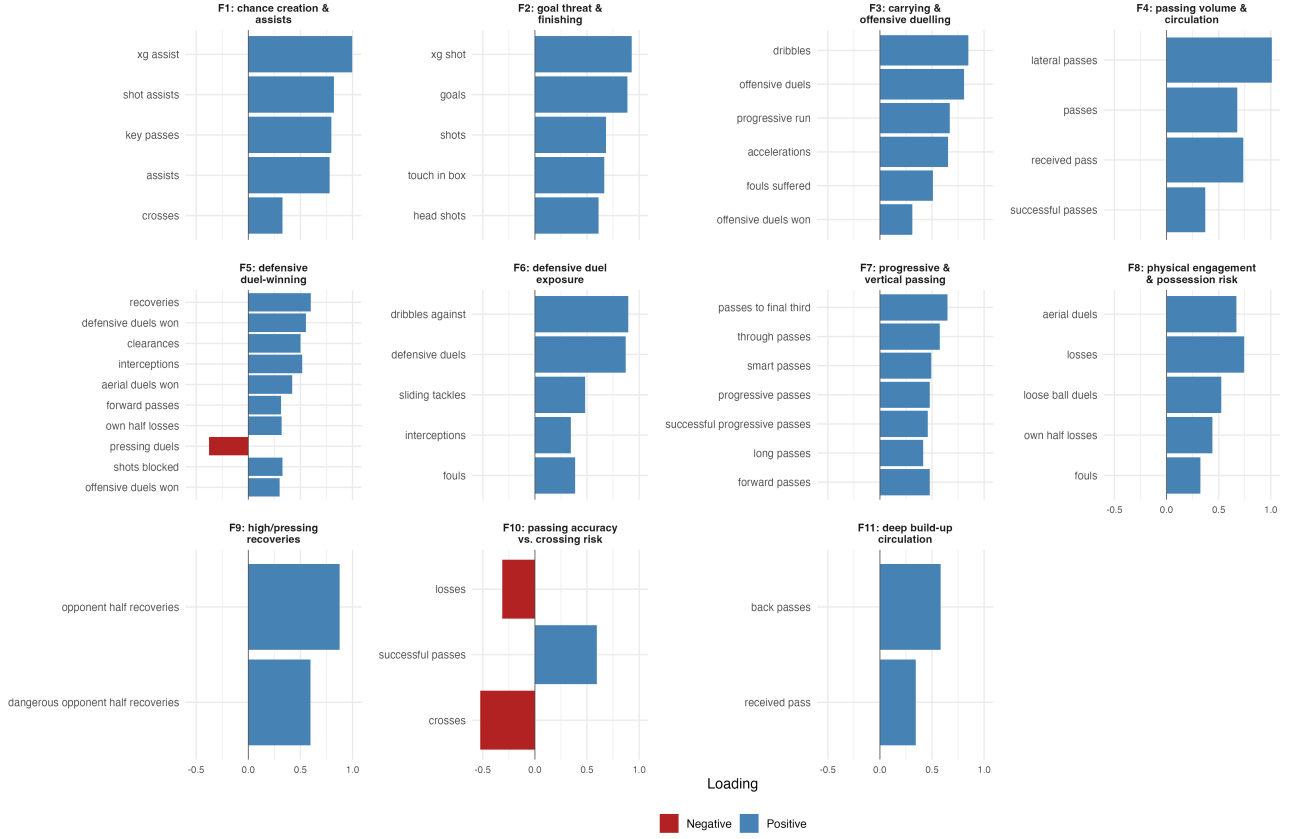


FIGURE 5. Rotated factor loadings for the eleven-factor oblimin solution, displaying variables with $|\text{loading}| \geq 0.30$. Blue and red bars indicate positive and negative loadings, respectively. Panels are ordered by variance explained (F1 highest) and labeled with the corresponding interpretation, matching the enumeration above.

be the centring matrix, and define

$$\mathbf{B} = -\frac{1}{2}\mathbf{J}\mathbf{D}^{(2)}\mathbf{J}.$$

Then $\mathbf{B}$ is the Gram matrix of the centred configuration. If

$$\mathbf{B} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$$

is its spectral decomposition, then the coordinates of the configuration can be recovered, up to rigid transformations, as

$$\mathbf{Z} = \mathbf{V}\mathbf{\Lambda}^{1/2}.$$

If only the first q positive eigenvalues are retained, so that

$$\mathbf{Z}_q = \mathbf{V}_q\mathbf{\Lambda}_q^{1/2},$$

then $\mathbf{Z}_q$ provides the best rank- q approximation to $\mathbf{B}$ in the least-squares sense given by the Eckart–Young theorem, and hence the classical q -dimensional CMDS representation obtained through spectral truncation [25, 7, 4].

Thus, the double-centred matrix provides a closed-form route to principal coordinates when the input dissimilarities are squared Euclidean distances. In that case, CMDS is essentially a distance-based route to the same geometry obtained from PCA on the centred data. When the input dissimilarities are not

squared Euclidean, as with cosine-based dissimilarity in the present exploratory analysis, the same spectral procedure gives an approximate Euclidean representation of that dissimilarity structure [7, 4]. In practice, the eigenvalue pattern of $\mathbf{B}$ is therefore informative about whether the distance structure is close to low-dimensional Euclidean and about how much information is lost when only a few dimensions are retained.

In the present work, CMDS was applied in particular to the cosine-based dissimilarity matrix. This matrix should be understood as an angular-based dissimilarity matrix rather than as a strict metric distance matrix. Cosine-based dissimilarity emphasizes the direction of the player profile rather than its overall norm, and when vectors are centred this directional comparison is equivalent to correlation. In this sense, two player–season observations may be regarded as similar not because they have the same absolute output, but because they exhibit a similar pattern across features. For example, two players with very different total event volumes may still be close under cosine-based dissimilarity if both display the same relative emphasis on progression, dribbling, defensive work, and duel involvement. CMDS then provides a first low-dimensional map of this pattern-based similarity structure.

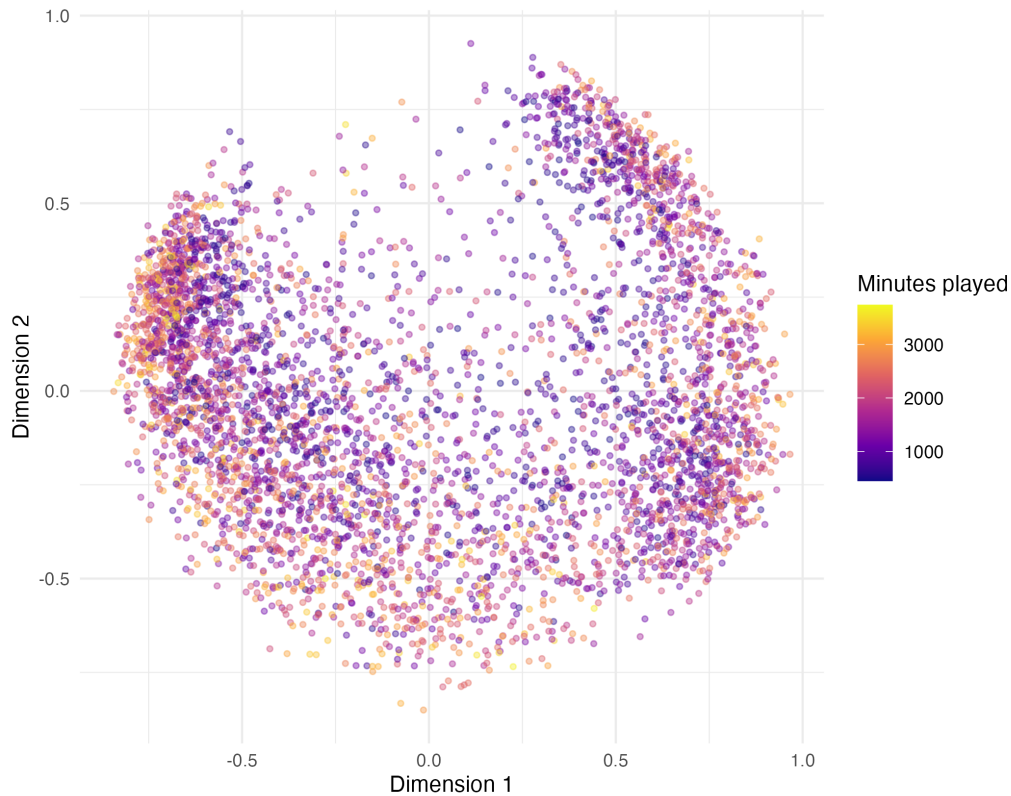


FIGURE 6. Two-dimensional CMDS embedding of the player–season observations using cosine-based dissimilarity. Points are coloured by minutes played. The absence of a clear minutes gradient suggests that the embedding is not primarily driven by playing time.

This representation is useful for exploratory purposes. It allows one to inspect whether the player–season observations occupy a coherent geometric space, whether certain profiles appear as extremes of the representation, and whether metadata-defined subsets of the data—for instance observations from different seasons, positional groups, or usage levels—occupy comparable regions of the map. At the same time, the limitations of CMDS are clear. It yields a deterministic embedding rather than a probabilistic one, provides no direct quantification of uncertainty for player locations, and does not by itself account for repeated observations of the same player across seasons or for more structured hierarchical variation.

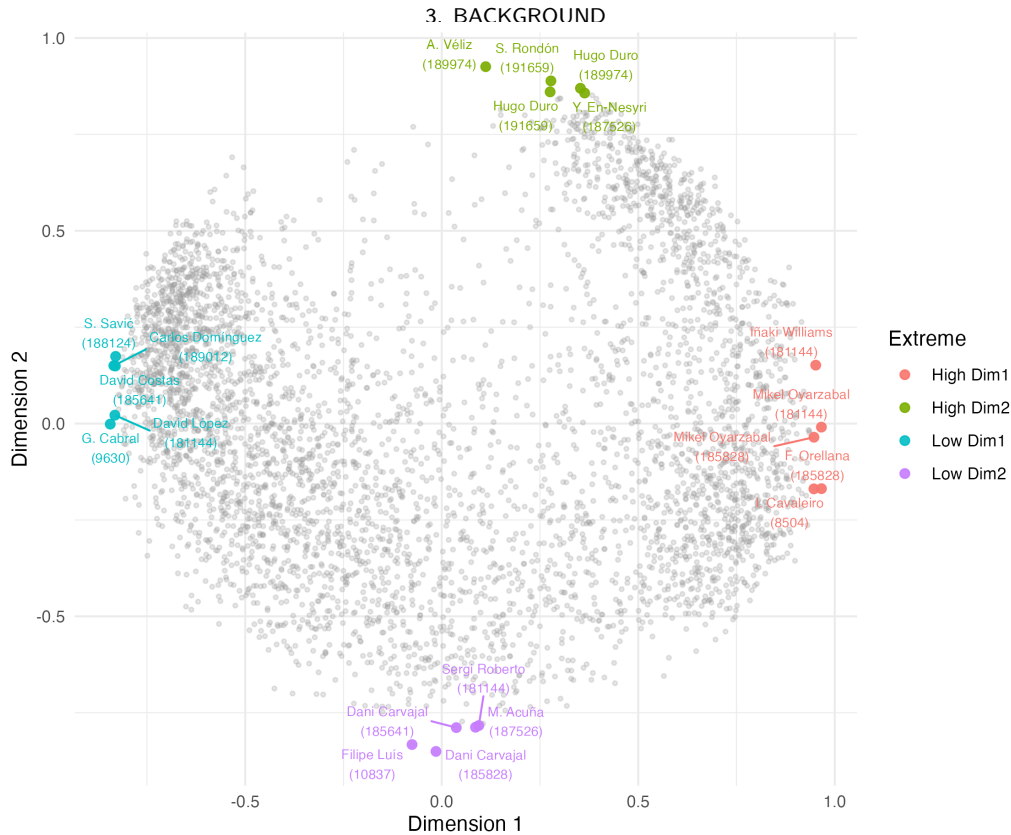


FIGURE 7. CMDS map with extreme observations labelled. The four groups of extremes correspond to high and low values along each displayed dimension. Dimension 1 is mainly anchored by wide attacking profiles on the positive side and centre-back profiles on the negative side, while Dimension 2 contrasts aerial or box-reference forwards with attacking full-back profiles.

The labelled extremes in Figure 7 help interpret the two displayed dimensions. The positive side of Dimension 1 is anchored mainly by wide attacking profiles, such as F. Orellana, Mikel Oyarzabal, Iñaki Williams, and I. Cavaleiro, whereas the negative side contains centre-back profiles, such as G. Cabral, Carlos Domínguez, David López, David Costas, and S. Savić. Dimension 2 separates aerial or penalty-box reference forwards, such as A. Véliz, S. Rondón, Hugo Duro, and Y. En-Nesyri, from attacking full-back profiles with substantial progression involvement, such as Dani Carvajal, Filipe Luís, M. Acuña, and Sergi Roberto. These labels should be interpreted descriptively: they identify the observed profiles anchoring the CMDS axes, not fixed universal football roles. For these reasons, CMDS is best viewed here as an exploratory geometric benchmark rather than as the final model. It indicates whether the chosen dissimilarity admits a coherent low-dimensional approximation, and it provides a descriptive precursor to more structured probabilistic latent-space models, including Bayesian and hierarchical extensions [18, 28].

2.4. Summary of the exploratory role of PCA, FA, and CMDS. Taken together, PCA, FA, and CMDS provide complementary descriptive evidence that the player–season data contain meaningful lower-dimensional structure. PCA suggests that a moderate number of directions captures a large proportion of total variation. FA indicates that several correlated latent traits may underlie the observed features. CMDS shows that the pairwise dissimilarity structure admits an interpretable low-dimensional Euclidean approximation.

These findings do not by themselves determine the final model, but they justify moving towards a latent-space formulation. More importantly, they reveal that the data contain enough structure for such a model to be meaningful, while also making clear that a purely descriptive approach is insufficient if one wishes to quantify uncertainty and account for the longitudinal character of player–season observations.

CHAPTER 4

Model

1. Motivation and overview

The exploratory analyses presented in Chapter 3 establish that the player–season feature space contains meaningful lower-dimensional structure. Principal component analysis revealed that approximately nineteen components are needed to account for ninety per cent of the total sample variance, while an eleven-factor oblimin solution provided an interpretable account of the empirical correlation structure through recognisable football dimensions such as goal threat, chance creation, pass circulation, defensive work, and duel efficiency. Classical multidimensional scaling applied to the cosine-based dissimilarity confirmed that the pairwise similarity structure admits a low-dimensional Euclidean approximation. Taken together, these findings support the view that player–season performance profiles are governed by a moderate number of shared latent traits, and that the feature space is rich enough for such traits to be recoverable from the data.

What the exploratory analyses cannot do, however, is quantify uncertainty in the latent structure. PCA scores and factor loadings are point estimates computed from the sample; they carry no natural notion of sampling variability, posterior distribution, or credible interval. This matters especially in a longitudinal setting, where the same player is observed in multiple seasons. The exploratory analyses treat all player–season rows as exchangeable, whereas a proper account of the data-generating process recognises that two observations belonging to the same player are not independent: they share the player’s stable latent characteristics. Ignoring this structure risks confusing player persistence with seasonal noise, and overstating the precision of latent-dimension estimates.

This chapter specifies a probabilistic model designed to address these limitations. The model is hierarchical: each player’s observed performance vector in a given season is the sum of three parts; a stable player effect, a season-specific shift, and residual noise. The player effects are themselves modelled as draws from a distribution whose covariance is structured as a low-rank factor model, directly encoding the finding from Chapter 3 that player-level variation is largely explained by a small number of shared latent dimensions.

The model has three principal objectives. First, it provides a principled variance decomposition, attributing each feature’s total variance to player, season, and residual components; this decomposition is expressed as a set of intraclass correlation coefficients, one per feature. Second, it estimates the covariance structure of the player effects through a factor model with K shared latent dimensions, recovering interpretable directions in the player-effect space. Third, all inferences are Bayesian: the model produces a joint posterior distribution over all parameters, from which uncertainty in the latent structure, the variance decomposition, and player-level factor scores is directly quantifiable [6].

2. Notation

Throughout this chapter, the following notation is used consistently. Let N denote the total number of player–season observations, I the number of unique players, S the number of seasons, P the number of standardised performance features, and K a positive integer referred to as the factor rank, whose selection will be discussed in Section 8. Observation indices run from $n = 1, \dots, N$; $i[n] \in \{1, \dots, I\}$ and $j[n] \in \{1, \dots, S\}$ are the player and season indices of observation n , respectively.

TABLE 1. Summary of notation used throughout this chapter.

Symbol	Meaning
N, I, S, P, K	Total observations; unique players; seasons; features; factor rank
$n = 1, \dots, N$	Observation index
$i[n] \in \{1, \dots, I\}$	Player index of observation n
$j[n] \in \{1, \dots, S\}$	Season index of observation n
$\mathbf{y}_n \in \mathbb{R}^P$	Standardised performance vector for observation n
$\mathbf{a}_i \in \mathbb{R}^P$	Player effect for player i
$\mathbf{b}_j \in \mathbb{R}^P$	Season effect for season j
$\varepsilon_n \in \mathbb{R}^P$	Residual for observation n
$\mathbf{\Lambda} \in \mathbb{R}^{P \times K}$	Factor loading matrix
$\mathbf{\Psi}_a = \text{diag}(\psi_{a,1}^2, \dots, \psi_{a,P}^2)$	Uniqueness matrix
$\mathbf{\Sigma}_a, \mathbf{\Sigma}_b, \mathbf{\Sigma}_e$	Player, season, and residual covariance matrices
$\boldsymbol{\eta}_i \in \mathbb{R}^K$	Latent factor scores for player i
$\mathbf{z}_{u,i} \in \mathbb{R}^P$	Latent uniqueness scores for player i
$\nu > 2$	Degrees-of-freedom parameter of the Student- t residual
$\phi \geq 0$	Observation-level variance scaling exponent
m_n	Minutes played in observation n
m_{ref}	Reference exposure level

3. The additive hierarchical decomposition

The central structural assumption of the model is that the performance vector for any given player–season observation arises from an additive decomposition into three independent latent components.

DEFINITION 4.1 (Additive decomposition). For each observation $n = 1, \dots, N$, the observed performance vector $\mathbf{y}_n \in \mathbb{R}^P$ satisfies

$$(1) \quad \mathbf{y}_n = \mathbf{a}_{i[n]} + \mathbf{b}_{j[n]} + \varepsilon_n,$$

where $\mathbf{a}_i \in \mathbb{R}^P$ is the *player effect* for player i , $\mathbf{b}_j \in \mathbb{R}^P$ is the *season effect* for season j , and $\varepsilon_n \in \mathbb{R}^P$ is the *residual* for observation n . The three components are assumed mutually independent.

The player effect $\mathbf{a}_i$ captures those aspects of player performance that are stable across seasons: the passing volume, defensive involvement, and goal-threat profile that characterise a player’s identity rather than a particular team context. The season effect $\mathbf{b}_j$ captures shifts in the distribution of performance statistics that are common to all players in season j : changes in the pace of play, officiating style, or data-collection conventions that may change the observed values of many features simultaneously. The residual ε_n captures what remains after both systematic components are removed, including player–season idiosyncrasy, noise from the finite number of matches played, and any tactical or physical changes specific to one season that are not representative of the player’s profile.

REMARK 4.2 (Absence of a global mean). No explicit population mean vector $\boldsymbol{\mu}$ appears in (1). This is because the features $\mathbf{y}_n$ are constructed by standardising each feature to zero mean and unit variance across the full sample, as described in Chapter 2. The population mean is therefore absorbed into the standardisation step, and the model is centred from the outset.

The decomposition (1) defines a *crossed* random-effects structure: players and seasons are crossed, not nested, each player appears in several seasons, and each season contains many players. This is in contrast to a nested structure in which seasons would be defined within players. The crossed structure is appropriate

here because the inferential goal is to separate player-level persistence from season-level context, not to model players as a random sample within fixed seasons [9].

4. Covariance structure

The three components $\mathbf{a}_i$, $\mathbf{b}_j$, and ε_n are each assigned a zero-mean multivariate distribution; they differ only in the structure of their covariance matrices. The choices here for said components differ: a richer covariance structure is assigned to the player effects, which are the primary object of interest and for which the data contain many observations per latent quantity, while simpler structures are assigned to season effects and residuals.

4.1. Player covariance: low-rank plus diagonal.

DEFINITION 4.3 (Factor covariance for player effects). The player effects $\mathbf{a}_i$ are exchangeable with zero mean and covariance

$$(2) \quad \Sigma_a = \mathbf{L}\mathbf{L}^\top + \Psi_a,$$

where $\mathbf{L} \in \mathbb{R}^{P \times K}$ is the loading matrix and $\Psi_a = \text{diag}(\psi_{a,1}^2, \dots, \psi_{a,P}^2)$ is the uniqueness matrix.

The structure in (2) decomposes the player-effect covariance into two parts. The term $\mathbf{L}\mathbf{L}^\top$ is a rank- K positive semi-definite matrix that captures the shared dependence among features through K latent dimensions; Ψ_a is a diagonal matrix that captures the feature-specific variance not explained by those shared dimensions.

PROPOSITION 4.4 (Entries of Σ_a). The diagonal entry $[\Sigma_a]_{pp}$ for feature p is

$$(3) \quad [\Sigma_a]_{pp} = \sum_{k=1}^K \Lambda_{pk}^2 + \psi_{a,p}^2,$$

and the off-diagonal entry for features $p \neq q$ is

$$(4) \quad [\Sigma_a]_{pq} = \sum_{k=1}^K \Lambda_{pk} \Lambda_{qk}.$$

PROOF. Both statements follow by expanding $[\mathbf{L}\mathbf{L}^\top + \Psi_a]_{pq}$. For the diagonal, $[\mathbf{L}\mathbf{L}^\top]_{pp} = \sum_k \Lambda_{pk}^2$ and $[\Psi_a]_{pp} = \psi_{a,p}^2$. For the off-diagonal, $[\mathbf{L}\mathbf{L}^\top]_{pq} = \sum_k \Lambda_{pk} \Lambda_{qk}$ and $[\Psi_a]_{pq} = 0$ since Ψ_a is diagonal. $\square$

In the factor-analysis literature, the quantity $\sum_{k=1}^K \Lambda_{pk}^2$ is called the *communality* of feature p : it measures how much of the player-level variance in feature p is explained by the K shared latent dimensions. The quantity $\psi_{a,p}^2$ is called the *uniqueness*: it is the portion of player-level variance in feature p that is specific to that feature and not captured by the shared structure. Together, they account for all player-level variance in feature p , as stated in (3).

The off-diagonal entries in (4) reflect the pairwise covariance between features p and q induced by the shared factors: two features co-vary at the player level only to the extent that they share common loadings on the K dimensions. A player who scores highly on a latent factor tends to score highly on all features with positive loadings on that factor, creating positive covariance among those features. The uniqueness matrix Ψ_a contributes nothing to off-diagonal entries, reflecting the assumption that, conditional on the K shared factors, feature-specific residuals are uncorrelated.

This covariance structure is motivated directly by the exploratory findings of Chapter 3. There, classical factor analysis showed that the 48 features are not independent at the player level: they cluster into correlated groups, and a number of shared latent traits (eleven, by parallel analysis) were needed to reproduce their correlation pattern.

The low-rank form $\Sigma_a = \Lambda\Lambda^\top + \Psi_a$ builds that same idea into the model's assumptions, rather than just describing it after the fact. It says exactly this: player-level covariance among features is mostly explained by K shared dimensions ($\Lambda\Lambda^\top$), plus a small, purely feature-specific leftover (Ψ_a).

The difference from the earlier exploratory chapter is that this is now a fully probabilistic (Bayesian) model, not a one-off descriptive result [17]. Classical factor analysis gives a single point-estimate set of loadings, with no sense of how uncertain they are. Here, Λ , Ψ_a , and everything built from them instead get a full posterior distribution: the model does not just claim that shared structure exists, it quantifies how confident it is about it. The rank K actually adopted for Σ_a is selected separately in Section 8, and Remark 4.19 explains why it need not, and does not, match the raw factor count above.

4.2. Season covariance: diagonal.

REMARK 4.5 (Diagonal season covariance). The season effects $\mathbf{b}_j$ are assigned zero mean and diagonal covariance:

$$(5) \quad \Sigma_b = \text{diag}(\sigma_{b,1}^2, \dots, \sigma_{b,P}^2).$$

This is an asymmetric design choice relative to the player covariance. A richer specification, $\Sigma_b = \Lambda_b\Lambda_b^\top + \Psi_b$, would model correlated season-level drift, but would require reliable estimation of $P \times K_b$ loading parameters from only S observations of the season. With $S = 12$ seasons, this is severely under-determined for any $K_b \geq 1$: preliminary fits with a low-rank Σ_b produced R-hat values exceeding 1.70 and bulk effective sample sizes as low as 6 for the season loadings, indicating that the posterior was entirely dominated by the prior. The diagonal specification (5) allows each feature its own season-level drift magnitude while avoiding parameters that the data cannot identify, plus the computational convenience.

4.3. Residual covariance. The residuals ε_n are assigned a diagonal covariance $\Sigma_e = \text{diag}(\sigma_{e,1}^2, \dots, \sigma_{e,P}^2)$, reflecting the assumption that, conditional on the player and season effects, the P features are uncorrelated at the observation level.

5. Likelihood specification

The additive decomposition (1) and the covariance structure of Section 4 are common to all model variants considered here. The models differ in the distributional form assigned to the scalar residuals ε_{np} , the p -th component of ε_n .

5.1. Normal baseline. The simplest complete specification assigns each residual component a Gaussian distribution:

$$(6) \quad \varepsilon_{np} \mid \sigma_{e,p} \sim \mathcal{N}(0, \sigma_{e,p}^2), \quad p = 1, \dots, P,$$

independently across features and observations, giving the conditional joint likelihood

$$(7) \quad \mathbf{y}_n \mid \mathbf{a}_{i[n]}, \mathbf{b}_{j[n]}, \sigma_e \sim \mathcal{N}_P(\mathbf{a}_{i[n]} + \mathbf{b}_{j[n]}, \Sigma_e).$$

This specification, referred to as the Normal model, serves as a computational and inferential baseline against which richer likelihood choices are compared.

5.2. Student- t residuals. For player–season data, the Gaussian assumption on residuals is restrictive. Some player–seasons produce extreme per-90 statistics for identifiable reasons: a player returning from injury may appear in only a handful of matches with atypical physical load; a player deployed in an unusual tactical role for one season may display statistics far from any stable average; a breakout season may represent an extreme but transient realisation. Under the Normal model, these extreme residuals exert a disproportionate influence on the estimates of the player and season effects, since the Gaussian density penalises departures from the conditional mean quadratically and without limit. A heavier-tailed distribution assigns such observations a higher residual likelihood and effectively down-weights their influence on the shared parameters.

DEFINITION 4.6 (Student- t error model). In the Student- t model, the residual for feature p in observation n follows

$$(8) \quad \varepsilon_{np} \mid \nu, \sigma_{e,p} \sim t(\nu, 0, \sigma_{e,p}), \quad p = 1, \dots, P,$$

independently across features and observations, where $\nu > 2$ is a degrees-of-freedom parameter shared across all features and observations, and $\sigma_{e,p} > 0$ is the feature-specific scale. The Student- t distribution with ν degrees of freedom, location zero, and scale σ has density

$$(9) \quad f(\varepsilon; \nu, \sigma) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2}) \sqrt{\nu\pi} \sigma} \left(1 + \frac{\varepsilon^2}{\nu\sigma^2}\right)^{-(\nu+1)/2}.$$

PROPOSITION 4.7 (Variance of the Student- t residual). Under Definition 4.6, the variance of ε_{np} is

$$(10) \quad \text{Var}(\varepsilon_{np}) = \sigma_{e,p}^2 \frac{\nu}{\nu - 2},$$

which is finite for $\nu > 2$ and converges to $\sigma_{e,p}^2$ as $\nu \rightarrow \infty$.

PROOF. The variance of a Student- t random variable with ν degrees of freedom and scale σ is $\sigma^2\nu/(\nu - 2)$, valid for $\nu > 2$. As $\nu \rightarrow \infty$, $\nu/(\nu - 2) \rightarrow 1$. $\square$

The tail-weight parameter ν has a direct interpretation: small values of ν produce heavier tails and greater likelihood of extreme residuals; large values recover the Gaussian behaviour [16, 10]. The degrees-of-freedom parameter is treated as unknown and estimated from the data; its posterior summarises the evidence in favour of heavy-tailed residuals conditional on the model specification and the prior.

5.3. Observation-level variance scaling. The residual scale $\sigma_{e,p}$ in (6) and (8) is constant across observations. This is a simplification. The performance features $\mathbf{y}_n$ are constructed as per-90 rates or efficiency ratios, both of which are ratio estimators: for a player who accumulated m_n minutes, the sampling variance of the per-90 estimate of feature p is approximately proportional to m_n^{-1} , independently of any structural player or season effects. An observation with 600 minutes of playing time is therefore inherently noisier than one with 2,500 minutes, even if both belong to players with identical latent profiles.

Failing to account for this heteroscedasticity may cause the model to attribute the elevated variance of low-minute observations to heavy residual tails (via a smaller ν), which would represent misspecification. A natural extension introduces a scaling of the residual variance on an observational level.

DEFINITION 4.8 (Minutes-weighted residual scale). Let $m_{\text{ref}} > 0$ denote a reference exposure level, taken as the sample median of the minutes-played values $\{m_n\}_{n=1}^N$. Define the observation-level residual standard deviation for observation n and feature p as

$$(11) \quad \sigma_{e,n,p} = \sigma_{e,p} \cdot \left(\frac{m_{\text{ref}}}{m_n}\right)^\phi,$$

where $\phi \geq 0$ is the scaling exponent. The Student- t model with minutes-weighting replaces $\sigma_{e,p}$ in (8) with $\sigma_{e,n,p}$.

Three canonical values of ϕ are immediately interpretable. Setting $\phi = 0$ recovers the base model (8), in which the residual scale is constant. Setting $\phi = 1/2$ corresponds to a square-root scaling analogous to the standard deviation of a Poisson count: variance grows linearly with event rate and standard deviation grows as the square root of inverse minutes. Setting $\phi = 1$ imposes full inverse-minutes scaling, so that the residual standard deviation is proportional to $m_n^{-1/2}$, as it would be for a simple average estimator under a constant-intensity Poisson process.

REMARK 4.9 (Fixed or estimated ϕ). The exponent ϕ may be handled in two ways. If ϕ is fixed at a pre-specified value ϕ_0 (for example, $\phi_0 = 1/2$), no additional parameters are introduced and the model remains as otherwise described; the choice of ϕ_0 encodes a prior judgement about the degree of heteroscedasticity due to playing time. If ϕ is treated as a free parameter, it is assigned a prior and sampled jointly with the remaining parameters. Identifiability of a free ϕ rests on the contrast between the residuals of high- and low-minute observations: given estimates of $\mathbf{a}_i$ and $\mathbf{b}_j$, the relationship between $|\varepsilon_{np}|$ and m_n is informative about ϕ . However, ϕ must be identified against $\sigma_{e,p}$ and ν simultaneously, since all three govern the scale and shape of the marginal residual distribution. Whether the fixed or estimated treatment is more appropriate for the present data is a diagnostic question addressed in Chapter 5.

6. Model identification

The factor model (2) introduces a structural non-identifiability in the loading matrix $\mathbf{\Lambda}$ that must be resolved before Bayesian inference is defined.

6.1. Rotational non-identifiability.

PROPOSITION 4.10 (Rotation equivalence). *For any orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{K \times K}$ satisfying $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}_K$, the loading matrices $\mathbf{\Lambda}$ and $\mathbf{\Lambda}\mathbf{Q}$ generate the same covariance:*

$$(12) \quad (\mathbf{\Lambda}\mathbf{Q})(\mathbf{\Lambda}\mathbf{Q})^\top = \mathbf{\Lambda}\mathbf{Q}\mathbf{Q}^\top\mathbf{\Lambda}^\top = \mathbf{\Lambda}\mathbf{\Lambda}^\top,$$

and consequently $(\mathbf{\Lambda}\mathbf{Q})(\mathbf{\Lambda}\mathbf{Q})^\top + \boldsymbol{\Psi}_a = \boldsymbol{\Sigma}_a$.

PROOF. Direct expansion using $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}_K$. □

Proposition 4.10 implies that the map $(\mathbf{\Lambda}, \boldsymbol{\Psi}_a) \mapsto \boldsymbol{\Sigma}_a$ is many-to-one: any element of the orbit $\{\mathbf{\Lambda}\mathbf{Q} : \mathbf{Q} \in O(K)\}$ produces the same covariance, where $O(K)$ denotes the group of $K \times K$ orthogonal matrices. The group $O(K)$ has dimension $K(K-1)/2$, corresponding to the degrees of freedom of a K -dimensional rotation; it also includes reflections, so that there are 2^K discrete sign symmetries in addition to the continuous rotational freedom. Without constraints, the posterior distribution over $\mathbf{\Lambda}$ is multimodal and the sampler cannot mix across symmetry-equivalent modes.

6.2. Lower-triangular positive-diagonal parametrisation. A classical and computationally convenient solution to the non-identifiability described above is to impose a constraint on $\mathbf{\Lambda}$ that selects exactly one representative from each equivalence class under $O(K)$.

DEFINITION 4.11 (LT-PD constraint). A matrix $\mathbf{\Lambda} \in \mathbb{R}^{P \times K}$ satisfies the *lower-triangular positive-diagonal* (LT-PD) constraint if:

- (i) $\Lambda_{pk} = 0$ for all $p < k$ (lower-triangular structure), and
- (ii) $\Lambda_{kk} > 0$ for all $k = 1, \dots, K$ (positive diagonal).

PROPOSITION 4.12 (Identification via LT-PD). *Suppose $P \geq K$ and let $\mathcal{L}$ denote the set of $P \times K$ matrices satisfying the LT-PD constraint. Then the map*

$$\mathcal{L} \times \{\boldsymbol{\Psi}_a : \psi_{a,p} > 0 \forall p\} \longrightarrow \mathcal{S}_P^+, \quad (\mathbf{\Lambda}, \boldsymbol{\Psi}_a) \mapsto \mathbf{\Lambda}\mathbf{\Lambda}^\top + \boldsymbol{\Psi}_a,$$

is injective, where $\mathcal{S}_P^+$ denotes the cone of $P \times P$ positive definite matrices. The factor covariance model is therefore identified under the LT-PD constraint.

The result in Proposition 4.12 is established in [1]; the lower-triangular structure eliminates the continuous rotational freedom by zeroing out the $K(K-1)/2$ upper-triangular entries, while the positive-diagonal constraint eliminates the 2^K discrete sign symmetries. An accessible treatment in the Bayesian factor analysis context is given in [11].

REMARK 4.13 (The LT-PD constraint is not an interpretive rotation). The LT-PD constraint resolves the mathematical non-identifiability [1, 11], but it does not produce loadings that are interpretable in a football sense. A simple-structure rotation such as varimax or oblimin minimises cross-loadings and maximises the separation of features across factors, yielding loading patterns that correspond to recognisable dimensions. The LT-PD constraint does neither: it fixes a coordinate system by zeroing out the upper triangle of $\mathbf{\Lambda}$, a choice determined by feature ordering rather than by football logic. The resulting columns of $\mathbf{\Lambda}$ do not correspond to canonical latent directions and should not be reported or interpreted as such. Interpretable factor directions are recovered through the post-processing procedure described in Section 10.

7. Prior specification

All model parameters are assigned priors that are weakly informative relative to the scale of the Z-standardised outcome $\mathbf{y}_n$. Because the features have unit variance by construction, a loading of magnitude $|\Lambda_{pk}| = 1$ represents a one-standard-deviation shift in feature p per unit change in the k -th factor score, which is already a large effect for a Z-scored performance variable. Priors concentrated near zero with tails extending to values of order one are therefore practically and scientifically reasonable.

DEFINITION 4.14 (Priors on the loading matrix). The diagonal entries of $\mathbf{\Lambda}$ are assigned

$$(13) \quad \Lambda_{kk} \sim \text{Half-Normal}(0, 1), \quad k = 1, \dots, K,$$

consistent with the LT-PD positivity constraint. The free lower-triangular entries are assigned

$$(14) \quad \Lambda_{pk} \sim \mathcal{N}(0, 1), \quad 1 \leq k < p \leq P.$$

DEFINITION 4.15 (Priors on variance parameters). The uniqueness standard deviations, the season standard deviations, and the residual standard deviations are each assigned independent Half-Normal priors:

$$(15) \quad \begin{aligned} \psi_{a,p} &\sim \text{Half-Normal}(0, 1), \\ \sigma_{b,p} &\sim \text{Half-Normal}(0, 1), \\ \sigma_{e,p} &\sim \text{Half-Normal}(0, 1), \quad p = 1, \dots, P. \end{aligned}$$

The Half-Normal(0, 1) prior for scale parameters concentrates most mass near zero while permitting values of order one with positive probability; it is recommended for variance components in hierarchical models as a weakly informative alternative to flat or improper priors, which can produce improper posteriors in hierarchical settings [8].

DEFINITION 4.16 (Prior on the degrees-of-freedom parameter). The Student- t degrees-of-freedom parameter is assigned

$$(16) \quad \nu \sim \text{Gamma}(2, 0.1),$$

supported on $\nu > 2$ to ensure finite residual variance. The Gamma(2, 0.1) distribution has mean 20 and standard deviation approximately 14. It places substantial mass below $\nu = 10$, heavy tails, while also allowing the posterior to concentrate near larger values if the data support behaviour close to Gaussian. The prior does not favour either the Gaussian limit $\nu \rightarrow \infty$ nor the extreme heavy-tail regime $\nu \rightarrow 2^+$ [15].

DEFINITION 4.17 (Prior on the scaling exponent). If the minutes-weighted residual scale of Definition 4.8 is used with ϕ treated as an unknown parameter, it is assigned

$$(17) \quad \phi \sim \text{Half-Normal}(0, 1),$$

which places most mass on moderate scaling effects ($\phi \lesssim 1$) while permitting stronger heteroscedasticity if supported by the data.

The joint prior is the product of all terms in Definitions 4.14–4.17, since all parameters are treated as independent then the joint prior density is:

$$\begin{aligned}
 p(\mathbf{\Lambda}, \psi_a, \sigma_b, \sigma_e, \nu, \phi) &= \prod_{k=1}^K p_{\text{HN}}(\Lambda_{kk}) \prod_{\substack{p=1 \\ p > k}}^P \mathcal{N}(\Lambda_{pk}; 0, 1) \\
 &\times \prod_{p=1}^P p_{\text{HN}}(\psi_{a,p}) p_{\text{HN}}(\sigma_{b,p}) p_{\text{HN}}(\sigma_{e,p}) \\
 &\times p_{\text{Ga}}(\nu) p_{\text{HN}}(\phi),
 \end{aligned}
 \tag{18}$$

where p_{HN} denotes the Half-Normal(0, 1) density and p_{Ga} denotes the Gamma(2, 0.1) density. The term involving ϕ is omitted when ϕ is held fixed.

8. Rank selection

The factor rank K governs the dimension of the latent space used to model the player covariance $\mathbf{\Sigma}_a$. Unlike the parameters $\mathbf{\Lambda}$, $\mathbf{\Psi}_a$, and ν , the rank K is not a continuous model parameter that can be assigned a prior and sampled within a single model: it determines the parametric family itself. The selection of K is therefore a model comparison problem.

For any fixed value of $K \in \{0, 1, 2, \dots\}$, the model specification in Sections 4–7 is complete. The case $K = 0$ is a special case: it sets $\mathbf{\Sigma}_a = \mathbf{\Psi}_a$ (diagonal player covariance), corresponding to the assumption that all player-level cross-feature correlation is zero. This *diagonal model* serves as a natural baseline; the gain in predictive performance relative to it quantifies the value of modelling shared player-level covariance structure.

Whether shared player-level covariance is worth modelling at all is answered decisively, and is reported in Chapter 5; the choice of rank among $K \geq 1$ is argued qualitatively, as follows.

8.1. A qualitative argument for the rank. One caveat is worth flagging briefly before making that argument: the rank $K^* = 3$ selected here is not expected to match the eleven-factor count found in Chapter 3’s exploratory analysis. The two describe different things, for reasons given in full in Remark 4.19 below.

The rank actually adopted for the production model is selected on qualitative, structural grounds instead: the adequacy of each fit’s own residuals, and the extent to which the shared player-effect covariance $\mathbf{\Sigma}_a$ is still adding well-identified structure as K increases. This mirrors standard practice in exploratory and confirmatory factor analysis, where the number of retained factors is routinely set by fit diagnostics and interpretability rather than by a single automated rule, and is consistent with the professional decision making process in my day-to-day job: that a qualitative argument, grounded in residuals and in the variance structure of $\mathbf{\Lambda}\mathbf{\Lambda}^\top$, is an appropriate and sufficient basis for closing the choice of K .

REMARK 4.18 (Anchoring to the posterior mean, not to per-draw estimates). Both criteria below require the eigendecomposition of the common-variance matrix $\mathbf{\Lambda}\mathbf{\Lambda}^\top$ introduced in Section 10. Proposition 4.23 shows that, *for a fixed posterior draw*, this eigendecomposition is invariant to the identification constraint used to fit $\mathbf{\Lambda}$. It is not, however, invariant *across* posterior draws in the sense needed to build a credible interval directly: sampling variability in $\mathbf{\Lambda}^{(s)}$ can rotate the leading eigenvectors within the subspace they span from one draw to the next, so naively comparing $\mathbf{V}^{(s)}$ across draws risks comparing different rotations of the same subspace rather than a stable estimate of one direction. To avoid this, the analyses below fix a single reference orientation—the eigendecomposition of the *posterior-mean* common-variance matrix $\bar{\mathbf{\Lambda}}\bar{\mathbf{\Lambda}}^\top$, where $\bar{\mathbf{\Lambda}} = \frac{1}{S_{\text{mc}}} \sum_s \mathbf{\Lambda}^{(s)}$ —and project each draw’s loadings onto this fixed reference before summarising uncertainty. The same construction gives comparable per-player credible intervals: replacing $\mathbf{V}^{(s)}$ by the

fixed $\bar{\mathbf{V}}$ in (25) yields $\mathbf{f}_i^{(s)} = \bar{\mathbf{V}}^\top \boldsymbol{\Lambda}^{(s)} \boldsymbol{\eta}_i^{(s)}$, used in Chapter 5 to report posterior credible intervals on individual players' canonical scores.

Criterion A: variance share of the PCs of $\bar{\mathbf{\Lambda}}\bar{\mathbf{\Lambda}}^\top$: For a candidate rank K , the eigenvalue d_K of the *last* (smallest) canonical factor, expressed as a share of $\sum_{k=1}^K d_k$, measures how much additional common-variance structure the K -th dimension contributes. A last-component share close to the trailing eigenvalues of the full uniqueness spectrum indicates that dimension is not resolving substantive shared structure.

Criterion B: residual adequacy under the fit's own $t(\hat{\nu})$ noise model: For each candidate fit, the standardised residuals $z_{np} = (y_{np} - \hat{a}_{i[n],p} - \hat{b}_{j[n],p}) / (\hat{\sigma}_{e,p} \sqrt{\hat{\nu}/(\hat{\nu} - 2)})$ are compared against the 99th-percentile threshold τ_{99} implied by $t(\hat{\nu})$. If the empirical exceedance rate across all $N \times P$ cells is at or below the nominal 1%, the residual distribution is adequate under the fitted noise model; this is a floor check on model misspecification, not a criterion that discriminates between two ranks that both pass it.

8.2. Selected rank. Applying Criteria A and B across $K = 2, 3, 4$ gives the following picture. The last canonical factor's common-variance share declines from 30.8% at $K = 2$ to 10.5% at $K = 3$ to 7.1% at $K = 4$ —a continued but decelerating decline, not a sharp drop to a noise floor at any point (Criterion A). Mean residual exceedance under τ_{99} is 0.74%, 0.77%, and 0.79% at $K = 2, 3, 4$ respectively—all comfortably below the 1% floor and essentially flat, so residual adequacy holds at every rank examined and does not discriminate among them (Criterion B).

Taken together, this comparison does not produce a clean reversal at $K = 4$ the way it does at $K = 3$ relative to $K = 2$: Criterion A mildly favours stopping at $K = 3$ (the fourth factor's contribution, while still non-trivial, is smaller than the third's), while Criterion B is uninformative at this rank range and offers no separate support either way. Read honestly, choosing $K = 3$ over $K = 4$ rests on Criterion A alone, and amounts to a preference for the simpler model: the fourth factor's contribution is smaller, but not small enough to ignore. This is a much weaker result than the case for $K \geq 1$ over $K = 0$, or even the milder case for $K = 3$ over $K = 2$, both of those were clear-cut; this one is not. Combined with the cross-validation trials described above, where $K = 4$ is likewise never clearly dominated, the honest summary is that the data do not sharply distinguish $K = 3$ from $K = 4$. $K = 3$ is nonetheless retained as the production rank $K^* = 3$, used throughout the remainder of this chapter and in Chapter 5: it is the more parsimonious of two ranks that the evidence does not clearly separate, and closing the model at a defensible, adequately-supported rank was judged preferable to an open-ended search for further, likely similarly ambiguous, comparisons at $K \geq 5$.

REMARK 4.19 (Comparability with the exploratory factor count of Chapter 3). Chapter 3 reports an eleven-factor solution from classical, maximum-likelihood factor analysis of the raw player–season feature matrix. It is natural to ask why the Bayesian model's rank, $K^* = 3$, is so much smaller. The two quantities are not estimates of the same object. The exploratory factor analysis of Chapter 3 is fit directly to the raw player–season observations $\mathbf{y}_n$, treated as exchangeable rows with no player/season structure: its factor count therefore describes the rank of the *total, unpartitioned* covariance of $\mathbf{y}_n$, in which any two features that co-vary for any reason shared stable player skill, a league-wide tactical shift common to a subset of seasons, or coincidental season-to-season drift—contribute jointly to the recovered factor structure. The additive decomposition of Definition 4.1, by contrast, first partitions this total covariance into player ($\boldsymbol{\Sigma}_a$), season ($\boldsymbol{\Sigma}_b$), and residual ($\boldsymbol{\Sigma}_e$) components, and the rank K studied in this section governs only $\boldsymbol{\Sigma}_a$: the covariance of what is stable about a player once season-level drift and observation-level noise have already been subtracted out. Since $\boldsymbol{\Sigma}_b$ is diagonal by construction (§4, justified there on identifiability grounds given only $S = 12$ seasons), any genuine cross-feature covariance that is specifically season-driven cannot be absorbed by $\boldsymbol{\Sigma}_b$ and is instead either left in the residual or folded into $\boldsymbol{\Sigma}_a$'s estimate; either way, K is not required to reproduce the rank of a covariance matrix that additionally contains season-level and

idiosyncratic sources of dependence. A smaller K for the player-only covariance than the raw factor count is therefore the expected outcome of conditioning on more structure, not evidence against either analysis. This reasoning is offered as an explanation for the direction of the discrepancy; it has not been separately validated (for instance, by fitting a raw factor analysis restricted to a player only, season averaged version of the data to check whether its factor count moves closer to three), and should be read as such.

9. Variance decomposition and intraclass correlation

A key inferential output of the model is a decomposition of the total variance of each feature into contributions from the player, season, and residual components. Under the additive model (1) and the mutual independence of $\mathbf{a}_i$, $\mathbf{b}_j$, and ε_n , the marginal variance of feature p in any observation is

$$(19) \quad \text{Var}(y_{np}) = [\boldsymbol{\Sigma}_a]_{pp} + \sigma_{b,p}^2 + \text{Var}(\varepsilon_{np}),$$

where $[\boldsymbol{\Sigma}_a]_{pp}$ is given by (3), and $\text{Var}(\varepsilon_{np}) = \sigma_{e,p}^2 \cdot \nu / (\nu - 2)$ by Proposition 4.7 for the Student- t model, or $\sigma_{e,p}^2$ for the Normal model.

DEFINITION 4.20 (Intraclass correlation coefficients). The player, season, and residual intraclass correlation coefficients for feature p are, respectively,

$$(20) \quad \text{ICC}_{a,p} = \frac{[\boldsymbol{\Sigma}_a]_{pp}}{[\boldsymbol{\Sigma}_a]_{pp} + \sigma_{b,p}^2 + \text{Var}(\varepsilon_{np})},$$

$$(21) \quad \text{ICC}_{b,p} = \frac{\sigma_{b,p}^2}{[\boldsymbol{\Sigma}_a]_{pp} + \sigma_{b,p}^2 + \text{Var}(\varepsilon_{np})},$$

$$(22) \quad \text{ICC}_{e,p} = \frac{\text{Var}(\varepsilon_{np})}{[\boldsymbol{\Sigma}_a]_{pp} + \sigma_{b,p}^2 + \text{Var}(\varepsilon_{np})},$$

satisfying $\text{ICC}_{a,p} + \text{ICC}_{b,p} + \text{ICC}_{e,p} = 1$ for all p .

The quantity $\text{ICC}_{a,p}$ measures the proportion of total variance in feature p that is attributable to stable differences among players. It is directly interpretable as the reliability of the feature as a measure of stable player style: features with $\text{ICC}_{a,p}$ close to one are dominated by stable player characteristics, while features with small $\text{ICC}_{a,p}$ are dominated by season-level context or observation-level noise. This framing of variance decomposition in terms of intraclass correlation follows [22].

REMARK 4.21 (Bayesian ICC). Because all variance parameters are sampled from the posterior, the quantities defined in Definition 4.20 are themselves posterior quantities. For each posterior draw, a value of $\text{ICC}_{a,p}$ can be computed from the sampled values of $\boldsymbol{\Lambda}$, $\boldsymbol{\Psi}_a$, $\sigma_{b,p}$, $\sigma_{e,p}$, and ν , yielding a full posterior distribution over each coefficient. Posterior medians and 95% credible intervals are reported in Chapter 5. The ICC framing is preferred over the raw variance decomposition because it is scale-free and comparable across features with different marginal variances.

10. Canonical factor directions via spectral decomposition

10.1. Why the identification constraint does not yield interpretable directions. The LT-PD constraint of Definition 4.11 resolves the rotational non-identifiability by fixing a particular coordinate system in the K -dimensional factor space, but it does so in a way that is mathematically convenient rather than interpretable. Specifically, the first column of $\boldsymbol{\Lambda}$ is constrained to load only on features 1 through P with feature 1 carrying the positive diagonal; the second column is constrained to load on features 2 through P with feature 2 carrying the positive diagonal; and so on. This triangular structure is a consequence of zeroing out the upper triangle of $\boldsymbol{\Lambda}$, not of any reasoning about football-domain structure. A different ordering of the P features would produce a different fitted $\boldsymbol{\Lambda}$, even though the resulting $\boldsymbol{\Sigma}_a$ would be unchanged.

The columns of the fitted LT-PD loading matrix therefore cannot be reported or interpreted as football archetypes.

10.2. Spectral decomposition of the common variance matrix. A rotation-invariant representation of the factor structure is obtained by working directly with the common variance matrix $\mathbf{C} = \mathbf{\Lambda}\mathbf{\Lambda}^\top$, which is uniquely determined by the model regardless of which identification constraint was used.

DEFINITION 4.22 (Spectral post-processing). For each posterior draw $s = 1, \dots, S_{\text{mc}}$, let $\mathbf{\Lambda}^{(s)}$ denote the sampled loading matrix. Compute the common variance matrix

$$(23) \quad \mathbf{C}^{(s)} = \mathbf{\Lambda}^{(s)} \left(\mathbf{\Lambda}^{(s)} \right)^\top \in \mathcal{S}_P^+,$$

and perform its spectral decomposition

$$(24) \quad \mathbf{C}^{(s)} = \mathbf{V}^{(s)} \mathbf{D}^{(s)} \left(\mathbf{V}^{(s)} \right)^\top,$$

where $\mathbf{D}^{(s)} = \text{diag}(d_1^{(s)} \geq \dots \geq d_K^{(s)} \geq 0)$ and $\mathbf{V}^{(s)} \in \mathbb{R}^{P \times K}$ has orthonormal columns. The k -th column $\mathbf{v}_k^{(s)}$ of $\mathbf{V}^{(s)}$ is the k -th principal eigenvector of $\mathbf{C}^{(s)}$; the corresponding eigenvalue $d_k^{(s)}$ is the amount of player-effect common variance explained by the k -th canonical factor direction. The rotated factor score for player i in draw s is

$$(25) \quad \mathbf{f}_i^{(s)} = \left(\mathbf{V}^{(s)} \right)^\top \boldsymbol{\eta}_i^{(s)} \in \mathbb{R}^K.$$

PROPOSITION 4.23 (Invariance of the spectral decomposition). *The eigenvalues $\{d_k^{(s)}\}_{k=1}^K$ and the common variance matrix $\mathbf{C}^{(s)}$ are invariant to any orthogonal transformation of the loading matrix. Replacing $\mathbf{\Lambda}^{(s)}$ by $\mathbf{\Lambda}^{(s)}\mathbf{Q}$ for any $\mathbf{Q} \in O(K)$ yields the same $\mathbf{C}^{(s)}$ and hence the same spectral decomposition (24).*

PROOF. By Proposition 4.10, $(\mathbf{\Lambda}^{(s)}\mathbf{Q})(\mathbf{\Lambda}^{(s)}\mathbf{Q})^\top = \mathbf{\Lambda}^{(s)}\mathbf{\Lambda}^{(s)\top} = \mathbf{C}^{(s)}$. $\square$

Proposition 4.23 is the key justification for reporting the eigenvectors $\mathbf{V}^{(s)}$ and eigenvalues $\{d_k^{(s)}\}$ as the canonical factor structure: their values depend only on the posterior common variance $\mathbf{C}^{(s)}$, not on which identification constraint was used to fit $\mathbf{\Lambda}$.

REMARK 4.24 (Sign convention). Eigenvectors are defined up to sign: if $\mathbf{v}$ is an eigenvector of $\mathbf{C}^{(s)}$, so is $-\mathbf{v}$. A sign convention is applied draw-by-draw before computing posterior summaries: for each canonical factor k , the sign of $\mathbf{v}_k^{(s)}$ is chosen so that the feature with the largest absolute loading on that eigenvector has a positive loading. This convention ensures that the posterior mean of $\mathbf{v}_k^{(s)}$ reflects a coherent direction rather than cancelling between draws of opposite sign.

CHAPTER 5

Results

This chapter reports the results of fitting the model of Chapter 4 to the production La Liga player–season dataset described in Chapter 2. Unless stated otherwise, every number, table, and figure in this chapter comes from the constant- σ_e , $K^* = 3$, Student- t production fit (Definition 4.1, Σ_a as in Definition 4.3) — the model whose specification, identification, and rank selection were argued for in Chapter 4. The one deliberate exception is §3, which resolves the fixed-vs-estimated ϕ question (Remark 4.9) using a pair of separate fits built specifically for that comparison.

1. Model diagnostics and convergence

Table 1 summarises posterior convergence for the production fit, grouped by parameter family. $\hat{R}$ and effective sample size are computed in the usual way from four parallel chains. Three of the 144 entries of Λ_a are excluded from the Λ_a row: under the lower-triangular, positive-diagonal constraint of §6.2 these entries are fixed at exactly zero and never sampled, so $\hat{R}$ is undefined for them (zero posterior variance) rather than poorly mixed.

TABLE 1. Posterior convergence summary for the production fit ($K^* = 3$), by parameter family. ν is the Student- t degrees of freedom (Definition 4.6); Λ_a , Ψ_a are the factor loadings and uniqueness (Definition 4.3); σ_b , σ_e are the season and residual scales. The $[\Lambda_a \Lambda_a^\top]_{1,10}$ row is the rotation-invariant bellwether check discussed below.

Parameter	N	ESS _{bulk} : median (min)	Max $\hat{R}$	$N(\hat{R} > 1.01)$
Λ_a	141	374 (130)	1.014	4
ν	1	2734 (2734)	1.000	0
ψ_a	48	1086 (528)	1.004	0
σ_b	48	1594 (1023)	1.005	0
σ_e	48	3422 (2210)	1.002	0
$[\Lambda_a \Lambda_a^\top]_{1,10}$	1	237 (237)	1.005	0

Convergence is excellent for the scalar and per-feature parameters: ν , ψ_a , σ_b , and σ_e all have $\hat{R} \leq 1.005$ and minimum bulk-ESS in the hundreds to thousands. Λ_a itself mixes less cleanly (worst $\hat{R} = 1.014$, 4 of the 141 sampled entries above the conventional 1.01 threshold) — but, as Remark 4.13 already anticipated, the raw loadings under the LT-PD identification constraint are an identification device, not the object of substantive interest, so this is not by itself informative about whether the model has recovered a genuine, stable shared-covariance structure. Proposition 4.23 gives the rotation-invariant object that *is* of substantive interest: the spectral decomposition of $\hat{\Lambda}_a \hat{\Lambda}_a^\top$. Following that proposition, mixing was checked directly on one representative entry of this rotation-invariant common-variance matrix, $[\Lambda_a \Lambda_a^\top]_{1,10}$ (the goals–forward-passes pair, previously the single worst-mixing entry identified during model development): $\hat{R} = 1.005$, bulk-ESS ≈ 237 , comfortably above the 200-draw threshold used throughout this chapter — a clean pass, in contrast to the raw loadings’ noisier picture.

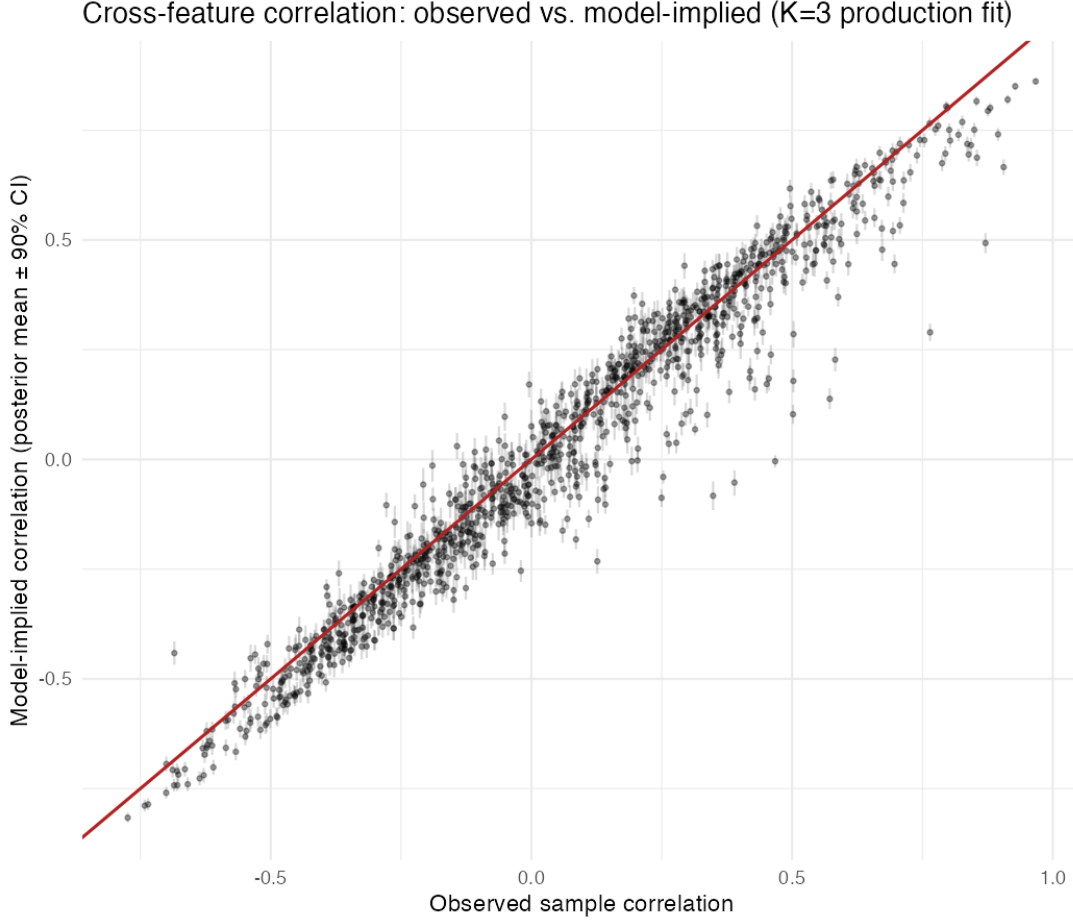


FIGURE 1. Posterior predictive check: observed pairwise feature correlations vs. the model-implied correlation $[\Sigma_a]_{pq} / \sqrt{\text{Var}(y_p)\text{Var}(y_q)}$ (Proposition 4.4), posterior mean $\pm 90\%$ credible interval across 4,000 draws. $r(\text{observed}, \text{model-implied}) = 0.975$.

Figure 1 checks whether the model reproduces the observed cross-feature correlation structure: since Σ_b and Σ_e are diagonal by construction, every model-implied off-diagonal correlation comes entirely from $\Sigma_a = \Lambda_a \Lambda_a^\top + \Psi_a$. The near-perfect agreement ($r = 0.975$) across all $\binom{48}{2} = 1,128$ feature pairs indicates the three shared factors, plus feature-specific uniqueness, are jointly sufficient to explain almost all of the pairwise dependence structure in the data — direct support for the low-rank modelling choice beyond the ELPD comparison of §2.

2. Rank selection: results

Two separate questions were posed in §8: whether low-rank Σ_a is worth having at all ($K = 0$ vs. $K \geq 1$), and, if so, which rank to use (K^* among $K \geq 1$). The evidence for the two questions is qualitatively different, and this section reports them separately rather than blending them.

$K = 0$ vs. $K \geq 1$: decisive. PSIS-LOO on the rebuilt production dataset ($N = 4,586$, $P = 48$) gives $\Delta\text{ELPD} = 5,820.2$ (SE 343.5, ≈ 17 standard errors) favouring the low-rank $K^* = 3$ model over the fully diagonal $K = 0$ baseline. Pareto- $\hat{k}$ exceeds 0.5 for the large majority of observations under both models (98.9% diagonal, 99.3% low-rank), so this magnitude should be read directionally rather than as a precise

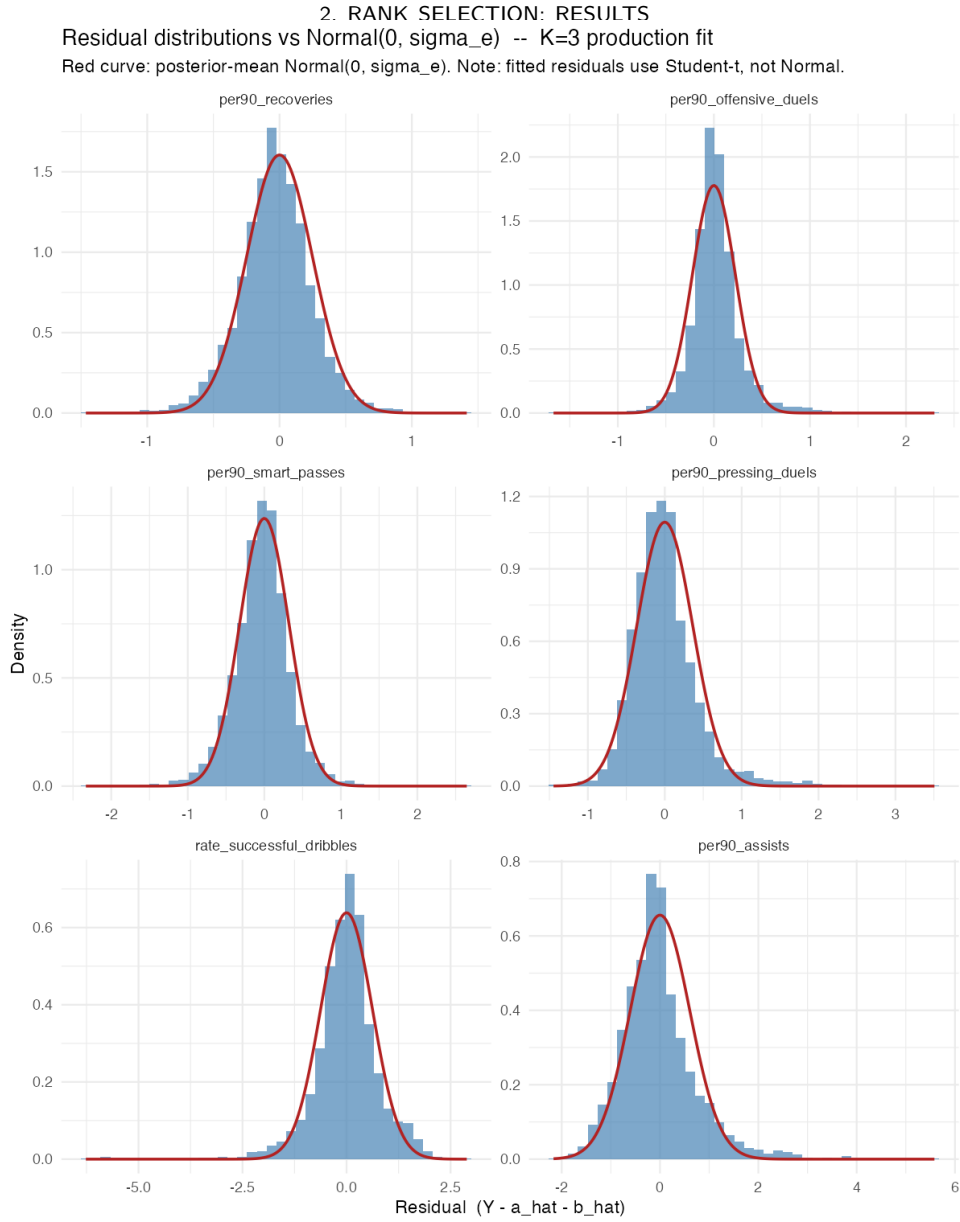


FIGURE 2. Residual check: $y_{np} - \hat{a}_{i[n],p} - \hat{b}_{j[n],p}$ against $\mathcal{N}(0, \hat{\sigma}_{e,p})$ for six features (top-2 by ICC_player, top-2 by ICC_season, bottom-2 by combined ICC; §4). The fitted noise model is Student- t , not Normal, so heavier-than-Normal tails here are expected, not a misspecification.

variance estimate — but a nearly-17-SE gap is not the kind of result a directional caveat can reverse. Shared player-level covariance structure is real and worth modelling.

Choosing K^* among $K \geq 1$: qualitative, per §8.1. Table 2 recaps the two structural criteria across $K = 2, 3, 4$ that §8.1 defined; a player-holdout cross-validation screen was also run across the same range and did not discriminate among these ranks either.

Criterion A mildly favours stopping at $K = 3$ (the fourth factor's 7.1% share is smaller than the third's 10.5%, but the decline from $K=2 \rightarrow 3 \rightarrow 4$ is continuous, not a sharp drop to a noise floor). Criterion

TABLE 2. Criteria A and B across candidate ranks (§8.1). Criterion A: last canonical factor's share of $\bar{\mathbf{A}}_a \bar{\mathbf{A}}_a^\top$'s total variance. Criterion B: mean fraction of standardised residual cells exceeding the fit's own $t(\hat{\nu})$ -implied τ_{99} threshold. The final column reports each fit's own estimated Student- t degrees of freedom $\hat{\nu}$, included for reference rather than as a third criterion.

K	Last-PC share of $\bar{\mathbf{A}}_a \bar{\mathbf{A}}_a^\top$	Mean $\Pr(z > \tau_{99})$	$\hat{\nu}$
2	30.8%	0.74%	4.88
3	10.5%	0.77%	4.90
4	7.1%	0.79%	4.95

B is flat and uninformative at this range (all three ranks comfortably under the 1% adequacy floor). Read honestly, the case for $K = 3$ over $K = 4$ rests on Criterion A alone, and amounts to a preference for the simpler model under a still-shrinking-but-not-negligible fourth factor, not a clean reversal of the kind that separates $K = 0$ from $K \geq 1$. $K^* = 3$ is retained on parsimony grounds and used throughout the rest of this chapter.

3. Resolving the observation-level variance scaling (ϕ)

Remark 4.9 left open whether the minutes-exposure scaling exponent ϕ in Definition 4.8 should be fixed or estimated, deferring the decision to this chapter. The motivation for scaling at all came from a direct diagnostic on the constant- σ_e fit: of the top-200 worst standardised-residual cells, 48.9% (88/180) came from the bottom quintile of playing time, against an expected 20% under no minutes effect (binomial $p < 4.4 \times 10^{-18}$) — observations from low-minute player-seasons are markedly under-fit by a model that treats residual variance as constant regardless of exposure.

TABLE 3. Resolving ϕ : constant- σ_e baseline vs. two minutes-scaled variants. "Low-minute worst-cell share" repeats the diagnostic above, recomputed on each fit's own residuals.

Model	ν (90% CI)	ϕ (90% CI)	Low-minute share
Constant σ_e (28)	4.90 [4.79, 5.01]	fixed at 0	48.9% (88/180)
$\phi = 0.5$ fixed (34a)	5.67 [5.54, 5.80]	fixed at 0.5	8.3% (14/168)
ϕ estimated (34b)	5.98 [5.83, 6.13]	0.366 [0.360, 0.373]	8.3% (14/168)

Table 3 compares the constant- σ_e production fit against two minutes-scaled refits: ϕ fixed at the Poisson-rate value of 0.5, and ϕ estimated freely. Both minutes-scaled variants eliminate the low-minute over-representation almost entirely (48.9% $\rightarrow$ 8.3%), confirming the scaling addresses the diagnosed problem rather than merely reshuffling it. The estimated $\hat{\phi} = 0.366$ (90% CI [0.360, 0.373]) is meaningfully below the fixed value of 0.5: the data reject pure Poisson-rate scaling in favour of a milder, super-Poisson exponent, consistent with some of a player's per-minute variance being genuine heterogeneity (skill, role) rather than pure sampling noise that would vanish with more minutes. Both scaled variants also push $\hat{\nu}$ higher (from 4.90 to 5.67–5.98): once exposure-driven heteroscedasticity is accounted for, less of the remaining spread needs to be attributed to heavy tails.

Verdict: minutes-scaling with an estimated ϕ is the better-supported treatment for this data. The rest of this chapter, however, continues to report results from the constant- σ_e production fit (stage 28): that is the fit all of the downstream infrastructure below (canonical loadings, archetypes, similarity) was built against, and it is what Chapter 4 describes as "the model" throughout. Re-deriving every subsequent result under the minutes-scaled variant is a natural extension, flagged here rather than undertaken, since

the qualitative substance of the loadings and similarity results is not expected to change materially and the constant- σ_e fit is the one already validated end-to-end.

4. Variance decomposition: intraclass correlations

Definition 4.20 defines $ICC_{a,p}$, $ICC_{b,p}$, $ICC_{e,p}$ as the share of feature p 's total variance attributable to player, season, and residual, respectively. Remark 4.21 notes these are themselves posterior quantities, since every variance component is sampled; Figure 3 plots the posterior-median split for the five features with the highest player ICC and, separately, the five features with the highest season ICC (the full 48-feature table is given in Table 1, Appendix A).

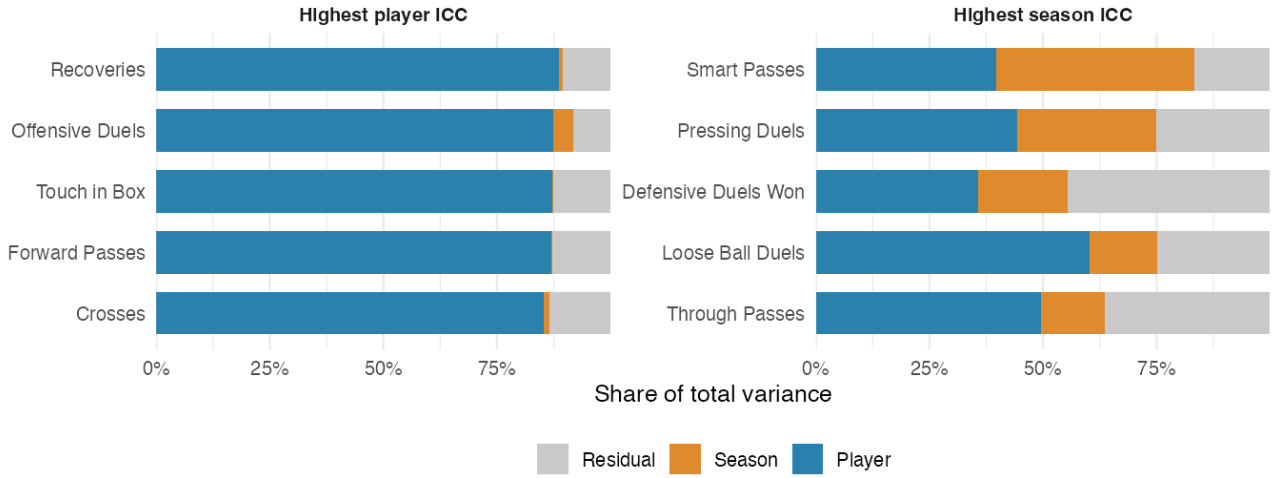


FIGURE 3. Player/season/residual variance share (posterior median) for the five features with the highest player ICC (left) and the five features with the highest season ICC (right).

Player identity dominates the left-panel features: all five exceed $ICC_a = 0.85$ (recoveries, offensive duels, touch in box, forward passes, crosses), with essentially no season component. The right panel tells a complementary story: season ICC reaches 0.44 for smart passes and 0.31 for pressing duels, features that do not otherwise stand out on the player-dominance ranking. This is the quantitative counterpart to the qualitative claim motivating the whole model (§4): football performance profiles are substantially stable properties of the player for most features, but a handful of specifically tactical or rule-sensitive features carry a real, non-trivial season signal — §7 returns to smart passes' season trend directly.

5. Canonical factor loadings by feature group

Raw Λ_a , fitted under the LT-PD constraint of §6.2, is an identification device and should not be interpreted directly (Remark 4.13). Following Definition 4.22, canonical factor directions are instead obtained by eigendecomposing the common-variance matrix $\bar{\Lambda}_a \bar{\Lambda}_a^\top$ (Proposition 4.23 shows this is invariant to the identification constraint used at fit time).

Roughly a quarter of total player-level variance sits in Ψ_a rather than the three shared factors (Table 4) — a meaningful idiosyncratic component that §8 returns to directly.

Three things stand out from Figure 4, one per axis.

PC1 separates defensive volume from attacking output. Defending and passing-volume features load positive — recoveries, interceptions, clearances, forward and long passes — while Attacking output loads entirely negative: all seven of its features, from goals to touches in the box, carry a negative, reliable loading.

TABLE 4. Variance shares of the three canonical factors, for the common-variance matrix $\bar{\mathbf{\Lambda}}_a \bar{\mathbf{\Lambda}}_a^\top$ and for total player variance $\bar{\mathbf{\Sigma}}_a = \bar{\mathbf{\Lambda}}_a \bar{\mathbf{\Lambda}}_a^\top + \bar{\mathbf{\Psi}}_a$. The gap between the two columns for each PC is the share absorbed by feature-specific uniqueness $\bar{\mathbf{\Psi}}_a$ instead of the shared factors.

Canonical factor	Share of $\bar{\mathbf{\Lambda}}_a \bar{\mathbf{\Lambda}}_a^\top$	Share of $\bar{\mathbf{\Sigma}}_a$
PC1	61.8%	45.6%
PC2	27.7%	20.7%
PC3	10.5%	8.1%

Passing itself splits by purpose rather than sitting on one side of the axis: bulk, directional passing (forward passes, long passes, lateral passes) loads with the defensive pole, while creative passing (key passes, smart passes, crosses, shot assists) loads with attacking output. This is the axis that separates centre-backs from strikers most cleanly: Table 5’s PC1 extremes are drawn entirely from those two positions, with no overlap between them.

PC2 separates technical, pass-based play from aerial and physical play. Passing volume, passing quality, and progressive actions load entirely positive; the negative pole is anchored by aerial duels and head shots, with clearances and blocked shots close behind and a mild negative loading on goals and expected goals — a target-man, aerial-duel profile rather than a purely defensive one. PC2 is largely independent of PC1: it is why a target forward and a deep-lying passer can sit at a similar point on the defend/attack axis while landing at opposite ends of this one, as Table 5’s PC2 rows show directly (Stuani, Budimir, Okazaki, and En-Nesyri at one extreme; Xavi, Iniesta, Rodríguez, and Kroos at the other).

PC3 does not match a simple wide-versus-direct-play description, and is renamed accordingly. Its positive pole combines possession quality — the successful-pass rate is the single largest loading on the axis — with direct goal threat: head shots, goals, and expected goals per shot all load positively alongside passing accuracy. Its negative pole is anchored by crossing, being dribbled past, and losing defensive duels, a mix of wide attacking play and defensive engagement rather than either one cleanly. PC3 adds information the other two axes do not carry: two attacking players who agree on PC1 and PC2 can still differ sharply in whether their output comes through retained possession and clean finishing or through crossing into contested situations, a distinction §8 returns to for two of its three case-study players.

Reconciling three factors with the earlier eleven. Chapter 3 reported an eleven-factor exploratory solution from classical factor analysis of the raw, unpartitioned player–season matrix. Remark 4.19 explains why $K^* = 3$ is not in tension with that finding: the classical analysis describes the total covariance of $\mathbf{y}_n$, in which any two features that co-vary for *any* reason — shared player skill, a league-wide tactical shift, or coincidental season drift — contribute jointly to the recovered factor count. $\bar{\mathbf{\Sigma}}_a$, by contrast, is the covariance of what is left of a player *after* season-level drift and observation noise are subtracted out (Definition 4.1); a smaller K for that narrower object is the expected consequence of conditioning on more structure, not a contradiction between the two analyses. As that remark itself notes, this explanation has not been independently validated (e.g. by refitting the classical FA on a player-only, season-averaged version of the data) and should be read as an argument for the direction of the discrepancy, not a demonstrated equivalence.

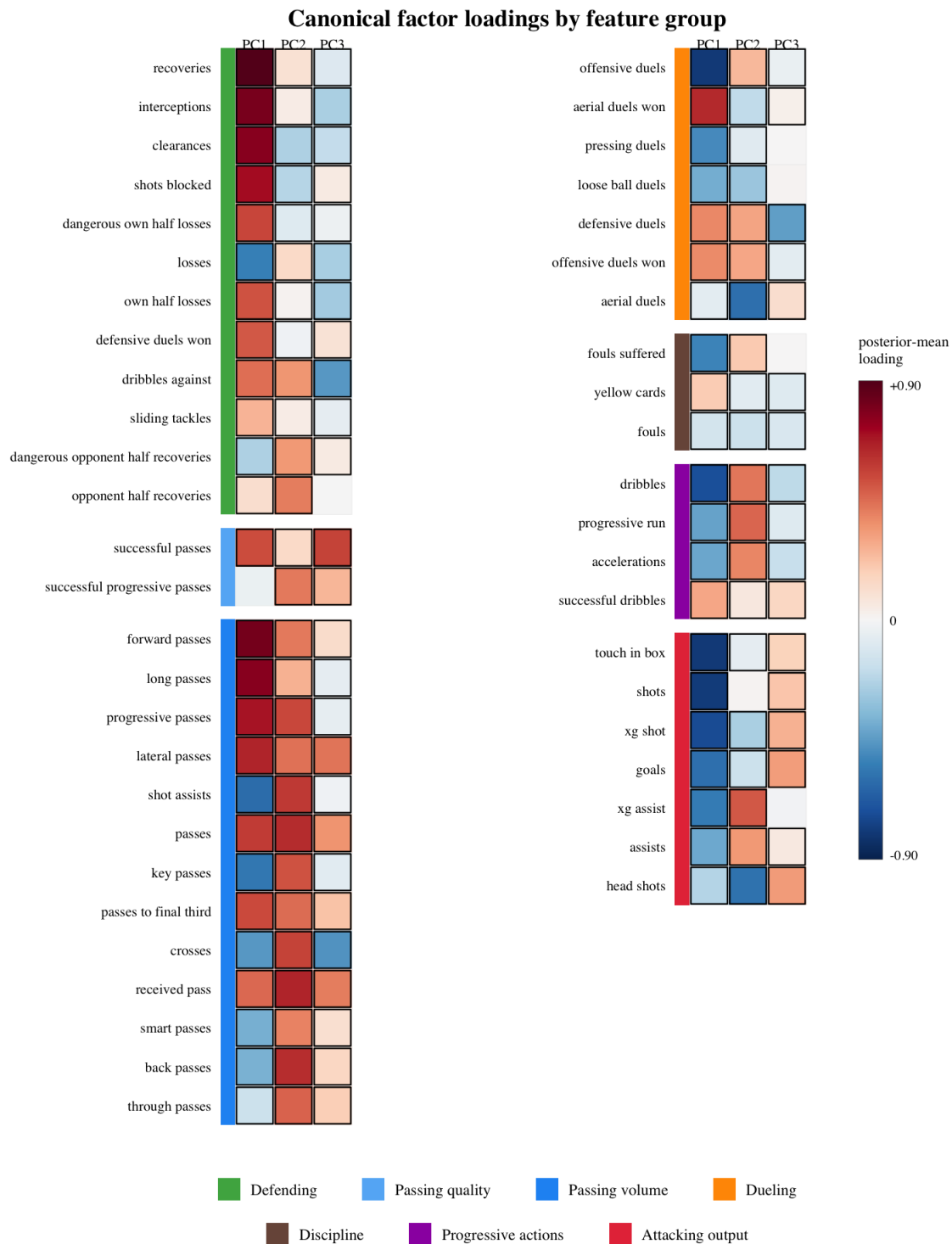


FIGURE 4. Posterior-mean loading of all 48 features on PC1–PC3, grouped by feature category and coloured on a diverging scale centred at zero. A dark outline marks the 137 of 144 cell-loadings (95%) whose 90% credible interval excludes zero; the seven unmarked cells are the exceptions. Groups are ordered so PC1’s sign runs monotonically from Defending (most positive) to Attacking output (most negative).

6. The canonical style space

Projecting each player's posterior-mean effect onto the three canonical directions gives every player a position in a 3-dimensional style space. k -means clustering ($k = 5$) on the (PC1, PC2) plane recovers five groups that align closely with recognisable positional roles once ordered by PC1: Forwards, Attacking midfielders, Midfielders, Defensive midfielders, and Centre backs, running from the most attacking to the most defensive end of PC1. This structure is not replotted below, since Figure 5 uses colour to track individual players across three specific projections rather than five population clusters, but it stands as independent confirmation that the space this chapter works with corresponds to real footballing structure, not an artefact of the model.

Canonical style space: case-study players (diamonds) and their three closest style-neighbours (circles)

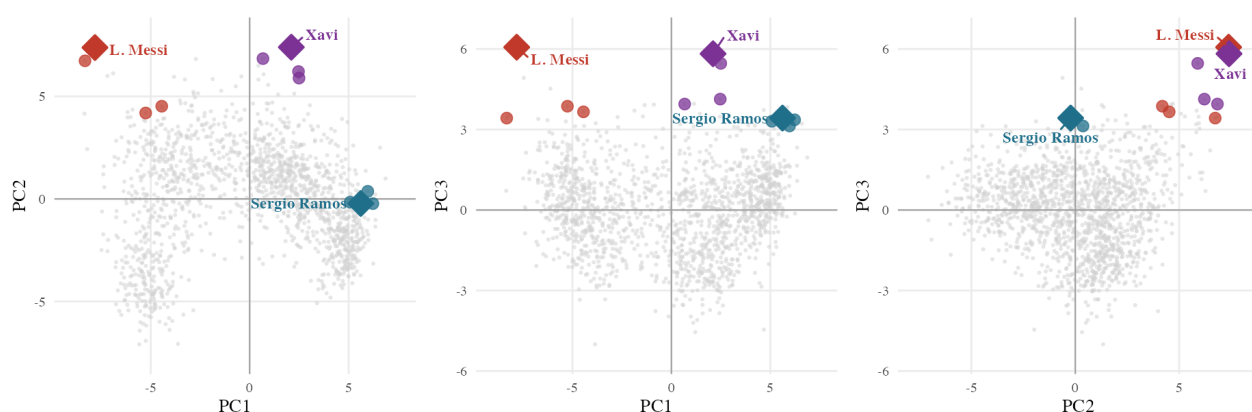


FIGURE 5. All 1,529 players' posterior-mean canonical scores (grey), with the three case-study players of §8 (diamonds) and their three closest style-neighbours (circles) in each of the three pairwise projections. Only the targets are labelled.

A three-dimensional style space is not honestly summarised by its first two coordinates, and Figure 5 shows why with the three players it highlights. On PC1 alone, Ramos is a clear defensive extreme, Messi a clear attacking one, and Xavi sits close to the population centre — a midfielder who is neither markedly defensive nor markedly attacking by this axis's measure. The PC1–PC2 panel adds the fact that Messi and Xavi both score highly on PC2, so a two-axis view already separates all three players cleanly. The PC2–PC3 panel tells a different story: Messi and Xavi's points sit almost on top of each other, both far to the technical, high-retention end of both axes, while Ramos remains clearly apart. Read on its own, that panel would suggest Messi and Xavi are close in style. Only PC1, absent from that projection, reveals that they are not — one of the more concrete illustrations in this chapter of why the third axis earns its place rather than being a minor correction to the first two.

Table 5 lists the most extreme players on each canonical factor, four per direction rather than eight, chosen as the highest-magnitude players on each axis who are also recognisable footballers (never reaching past roughly the top 15–20 by magnitude to find one). The one exception is PC3's bottom row: its highest-magnitude players on that (negative) pole are all genuinely unfamiliar names, so the row is left as the honest, unadjusted top 4 by magnitude rather than reaching deep into the ranking for recognisability.

TABLE 5. Top and bottom 4 players on each canonical factor (PC1: defensive volume vs. attacking output; PC2: technical/pass-based vs. aerial/physical; PC3: ball retention and direct scoring vs. crossing and defensive engagement).

PC1: defensive vs. attacking		PC2: technical vs. aerial		PC3: retention vs. crossing	
Top					
J. Mascherano	6.59	Xavi	7.40	L. Messi	6.07
Gerard Piqué	6.23	Andrés Iniesta	6.84	Xavi	5.82
S. Umtiti	5.97	J. Rodríguez	6.82	M. Pjanić	5.46
Raúl Albiol	5.80	T. Kroos	6.21	Cristiano Ronaldo	4.92
Bottom					
Vinícius Júnior	-8.35	C. Stuani	-7.09	Iván Alejo	-5.00
Neymar	-8.32	A. Budimir	-6.43	Juan Iglesias	-4.55
K. Mbappé	-8.21	S. Okazaki	-6.11	O. Vranješ	-4.46
L. Messi	-7.82	Y. En-Nesyri	-6.09	Houboulang Mendes	-4.20

7. Season profiles

Consistent with Figure 3, season ICC is small for almost every feature — but small does not mean zero, and group-averaged season effects reveal coherent league-wide drift over the twelve seasons in the data (Figure 6).

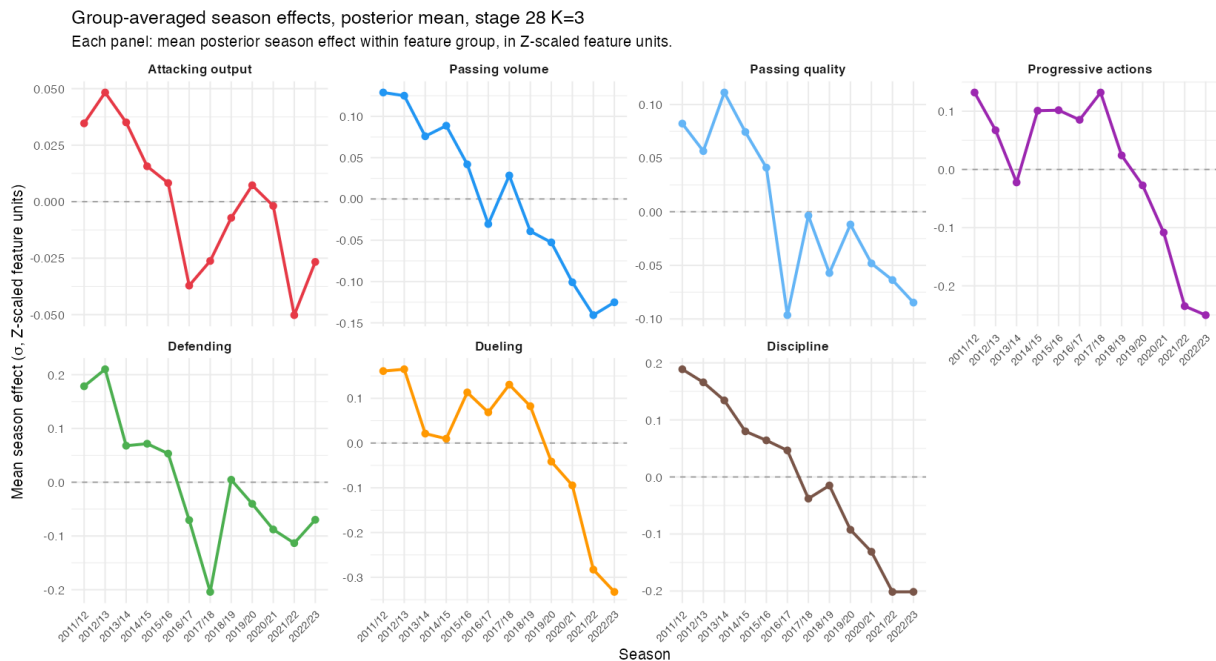


FIGURE 6. Group-averaged posterior-mean season effects $\hat{\mathbf{b}}_j$, in Z-scaled feature units, one panel per feature group, 2011/12–2022/23. Most groups drift downward from around 2017/18 onward.

Smart passes stands out not just for its range but for its shape: every other feature in Figure 7 rises, falls, and partially recovers within the sample window, while smart passes falls steadily from 2011/12 to

Top 6 features by season-effect range

Posterior mean season effect $\hat{b}_j$ with 90% credible intervals. Range = max – min of posterior mean across 12 La Liga seasons (2011/12–2022/23).



FIGURE 7. The six features with the largest peak-to-trough range in $\hat{b}_j$ across seasons. Smart passes shows the largest range (2.4σ) and, uniquely among the six, a monotonic falling trend rather than a peak-and-recovery shape.

2022/23 with no recovery at any point. A through-ball attempted at a league-average rate in 2011/12 would be attempted well below that rate a decade later, holding player identity fixed. Read as a tactical signal rather than a statistical curiosity, this is evidence of a leaguwide shift away from risk-laden, line-breaking passing and toward lower-risk possession retention over the sample window — consistent with the broader decline visible in the Passing volume and Passing quality panels of Figure 6.

8. Player similarity: separating style from level

This section takes up, in depth, the question the Introduction posed at the outset: what does it mean for two players to be “similar,” and can a player be stylistically similar to an outlier without matching their output?

8.1. Motivation: separating style from level. The model’s own covariance structure already gives a principled way to make this split. Recall (Definition 4.3) that the player effect decomposes as

$$\mathbf{a}_i = \mathbf{\Lambda}_a \boldsymbol{\eta}_i + \boldsymbol{\psi}_a \odot \mathbf{z}_{u,i},$$

a *shared* component driven by the $K^* = 3$ canonical factors, and an *idiosyncratic* component that those factors do not explain. **Style** is naturally read off the shared component: a player's position in the 3-dimensional canonical factor space, $\mathbf{f}_i$ (Definition 4.22), the same coordinates used for the archetype clustering of §6. **Level** is naturally read off the specific output features that shared style position does not by itself pin down — here, goals and expected goals per shot. Both are evaluated at their posterior mean throughout this section.

Two players can therefore be *style-close* (near-identical $\mathbf{f}_i$) while being *level-far* (very different modelled goal output). That is the model-native version of “can a player be stylistically similar to an outlier without matching their production”: yes, and the mechanism is explicit rather than asserted. It also explains why a true generational outlier is unlikely to have a perfect twin — their outlier-ness lives disproportionately in $\psi_a \odot \mathbf{z}_{u,i}$, the part of their identity that is, by construction, not shared with anyone else. Matching someone's shared style is achievable; matching both their style and their most idiosyncratic output is structurally much less likely.

A note on scope: the dataset is La Liga only, 2011/12–2022/23 (Chapter 2). A player like Erling Haaland, who has never played in this league, is not in this analysis. He is a useful illustration of the “generational outlier” idea in the abstract, never a literal query target here — every concrete example below uses an actual in-sample player as the stand-in.

8.2. Case studies: three targets. Style distance between two players is the Euclidean distance between their posterior-mean canonical factor scores $\bar{\mathbf{f}}_i$, standardised across all 1,529 outfield players; level gap is the Euclidean distance between their posterior-mean per90 goals and expected-goals effects. Both are single numbers per pair, reported directly as distances rather than converted to a bounded similarity score: smaller means more similar, and the underlying quantity is already a Euclidean distance in standardised canonical-score space, the same one used throughout this chapter. Introducing a monotone transform to a $[0, 1]$ scale here would add a mathematical object the thesis does not otherwise need.

The remainder of this section works through three named players: Messi, Sergio Ramos, and Xavi. The primary reason for this choice is that a reader already has an informed opinion about how each of these three plays, which is what makes the model's output checkable rather than merely plausible — an unfamiliar player's neighbour list cannot be judged against prior knowledge, a familiar one's can. A secondary reason is that their canonical scores span real extremes of the space, though less completely than a one-player-per-axis account would suggest. Messi is an extreme on both PC1 (4th most extreme of 1,529 players) and PC3 (the single most extreme value in the dataset). Xavi is an extreme on PC2 (also the single most extreme value) and PC3 (2nd most extreme), but sits close to the population centre on PC1 (1,123rd by magnitude, near the median). Ramos is a clear PC1 extreme (26th most extreme, the positive pole) but not a PC2 extreme (1,451st, essentially the population median) and shares PC3's negative pole with Messi and Xavi rather than anchoring an opposing one. Between the three, both poles of PC1 are covered and the shared negative pole of PC2 and PC3 is covered twice over; PC2's and PC3's positive poles are not represented by any of the three. This is a genuine partial coverage of the space, not a tidy one-axis-each account, and is reported as such.

The three most distant players per target are chosen from the top decile of style distance from that target, as the three most recognisable names in that decile, capped at not reaching past roughly its top 30 players by raw distance; the rule is applied identically to all three targets. For Ramos and Xavi this returns genuinely recognisable players (Hélder Costa, Messi, and Pedro Porro for Ramos; Ante Budimir, Cristian Stuani, and Youssef En-Nesyri for Xavi). For Messi it does not: no player in his top decile of style distance is a recognisable name at any reasonable depth, so the honest top three by raw distance are used instead — O. Vranješ, Juan Iglesias, and Iván Alejo, all low-minute fringe players. That the players stylistically furthest from Messi are unheralded role-players rather than other attacking talents is itself informative: his profile is extreme enough that its opposite is occupied by nobody in particular.

TABLE 6. Each target's three closest style-neighbours, by style distance, with the level gap between them.

Target	Similar player	Style distance	Level gap
L. Messi	Neymar	1.736	1.988
	Pedro	2.014	2.757
	Philippe Coutinho	2.115	3.135
Sergio Ramos	Javi García	0.161	1.126
	Gerard Piqué	0.164	0.772
	S. Umtiti	0.316	1.330
Xavi	M. Pjanić	0.646	0.321
	T. Kroos	1.195	0.438
	Andrés Iniesta	1.289	0.620

Messi. Neymar, Pedro, and Philippe Coutinho are Messi's three closest style-neighbours, and none closes the level gap: Neymar's, the smallest of the three, is still 1.99. Figure 8's PC1 and PC3 intervals for these three overlap Messi's own comfortably, while the three most distant players sit clearly apart on every axis.

Ramos. Javi García (0.161) and Gerard Piqué (0.164) sit within 0.003 of each other in style distance from Ramos, both clearly ahead of the third-closest, Samuel Umtiti (0.316). Figure 8 shows what the point estimates alone do not: on PC1, Ramos's own credible interval overlaps all three neighbours, and García's and Piqué's overlap each other almost entirely. The near-tie in Table 6 is a real near-tie in the posterior, not an artefact of reporting three decimal places. Piqué's level gap to Ramos (0.77) is smaller than García's (1.13), which may explain why the well-known Ramos–Piqué pairing persists in football conversation even though García edges ahead on style alone. One of Ramos's three most distant players is Messi himself — the same PC1 contrast this chapter returns to repeatedly, restated here as a distance rather than a pair of separate extremes.

Xavi. Miralem Pjanić (0.65), Toni Kroos (1.19), and Andrés Iniesta (1.29) are three players built around retaining and circulating possession, matching PC2's description of Xavi's game directly, and all three level gaps are small (Pjanić's, 0.32, is the smallest anywhere in Table 6). Xavi's three most distant players — Budimir, Stuani, and En-Nesyri — are all aerial-target forwards, PC2's opposite pole. Figure 8's PC2 panel makes the contrast visually explicit: Xavi's cluster and the three forwards' do not overlap at all.

Canonical factor scores with 90% credible intervals: each target (diamond), its three closest style-neighbours (circles), and the three most distant players (grey squares)

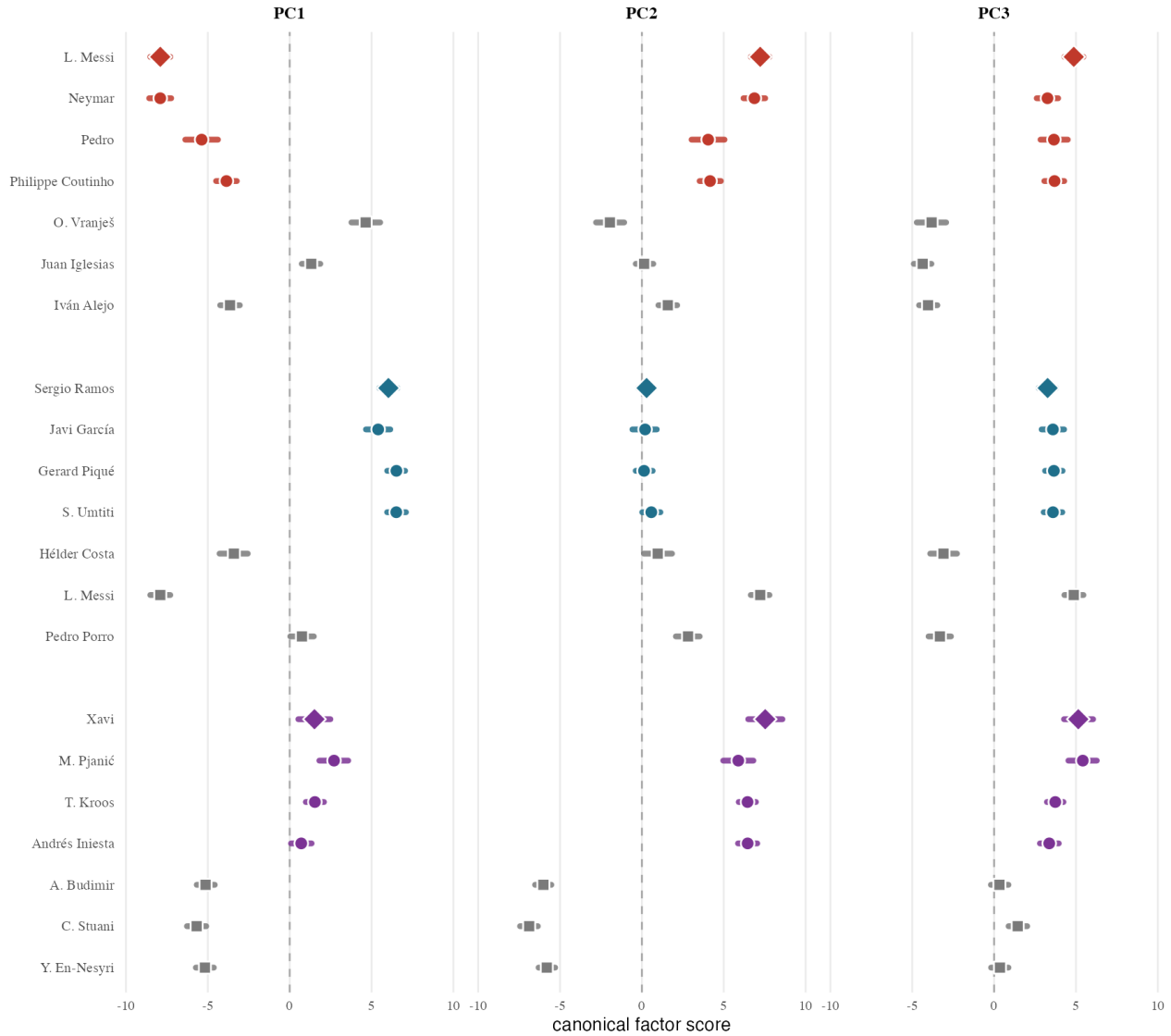


FIGURE 8. Posterior 90% credible intervals for each target's canonical score on PC1–PC3, alongside the same three style-neighbours as Table 6 and the three most distant players by style (grey squares; selection rule in the text). Scores use the fixed reference orientation of Remark 4.18, so intervals are directly comparable across players and draws.

9. Applications: scouting and recommendations

The style/level decomposition of §8 turns directly into a scouting-relevant product: a shortlist of players who match a target's style at a different output tier, rather than an attempted like-for-like replacement of an irreplaceable talent. Concretely, the recipe is: fix a target player, take their top style-neighbours (as in Table 6), and read off which of those neighbours has the largest *level gap* — that neighbour is a plausible “cheaper alternative” candidate, in the sense of matching shape while differing in the specific output dimension a club may be more willing to accept a discount on.

This is a style/production tool, not a recruitment decision system, and should be read as one input among many: market value, age, injury history, contract situation, and league context are not modelled anywhere in this pipeline. Within those limits, it offers something a purely descriptive scouting report cannot: an explicit separation of “how a player plays” from “how much they currently produce.”

10. Future work

Three directions were considered during this work and are reported here as scoped future work rather than as contributions of the present thesis.

Uncertainty beyond posterior means. The model itself is fully Bayesian throughout: every variance component, loading, and rank-selection criterion in Chapters 4 and 5 carries a posterior distribution, not a point estimate, and individual canonical scores are no longer an exception — Figure 8 reports a 90% credible interval for each case-study player on every axis, under the fixed reference orientation of Remark 4.18. Two other summaries still reduce the posterior to a single number. The style-space clustering of §6 assigns each player to a single k -means cluster on $\bar{\mathbf{f}}_i$, and the similarity table of §8 reports one style distance and one level gap per pair rather than a credible interval on either. A soft, probability-weighted archetype membership per player (e.g. from a Gaussian mixture over the canonical factor scores, letting a player be reported as, say, 70% centre-back and 30% defensive midfielder) is a direct extension of the scores now being reported with uncertainty, and remains open. A credible interval on a *pairwise distance* is a different, harder problem: it depends on the joint draw-by-draw behaviour of two players’ scores together, not on either player’s marginal interval alone, and was not attempted here.

Distance-based latent-space models. The factor-analytic route taken in Chapter 4 represents player style through a shared covariance structure. A complementary approach, not pursued here, represents player style directly through pairwise dissimilarities via Bayesian multidimensional scaling [18], with hierarchical extensions available for repeated player–season observations [28]. The classical multidimensional scaling analysis of Chapter 3 is a first, non-probabilistic step in that direction; a fully Bayesian version would attach posterior uncertainty directly to each player’s position in the similarity space, which the present factor-analytic model instead obtains indirectly, through the canonical factor scores.

Team-season effects and tactical fit. No team identifier exists anywhere in the current data pipeline (confirmed absent from both the raw SQL extraction and the model’s row-mapping table), so a genuine player-vs-team-system comparison is not buildable without new data engineering back to the Wyscout source. Adding a team-season effect $c_{t(i,j),j}$ would separate manager and tactical-system effects from stable player identity, and would be the natural foundation for a real tactical-fit score; absent that extension, the closest available proxy is simply which of the five archetype clusters (§6) a player belongs to.

APPENDIX A

Full intraclass-correlation table

The full 48-feature player, season, and residual ICC table referenced from §4 is given below (posterior median with 90% credible interval), ordered by descending player ICC; Figure 3 highlights the ten most informative rows directly in the main text.

Table 1. Full player, season, and residual ICC (posterior median [90% CI]) for all 48 features, ordered by descending player ICC.

Feature	ICC(player)	ICC(season)	ICC(resid.)
recoveries	0.89 [0.88, 0.90]	0.01 [0.00, 0.02]	0.11 [0.10, 0.11]
offensive duels	0.87 [0.83, 0.90]	0.04 [0.02, 0.09]	0.08 [0.08, 0.09]
touch in box	0.87 [0.86, 0.88]	0.00 [0.00, 0.00]	0.13 [0.12, 0.14]
forward passes	0.87 [0.86, 0.88]	0.00 [0.00, 0.01]	0.13 [0.12, 0.14]
crosses	0.85 [0.84, 0.86]	0.01 [0.01, 0.03]	0.13 [0.12, 0.14]
passes	0.85 [0.83, 0.86]	0.01 [0.00, 0.02]	0.14 [0.13, 0.15]
long passes	0.84 [0.82, 0.85]	0.00 [0.00, 0.01]	0.16 [0.15, 0.17]
xg shot	0.83 [0.82, 0.84]	0.00 [0.00, 0.00]	0.17 [0.16, 0.18]
clearances	0.83 [0.81, 0.85]	0.02 [0.01, 0.04]	0.15 [0.14, 0.16]
shots	0.83 [0.81, 0.84]	0.01 [0.00, 0.02]	0.17 [0.16, 0.18]
lateral passes	0.82 [0.80, 0.83]	0.01 [0.01, 0.03]	0.17 [0.16, 0.18]
progressive passes	0.81 [0.77, 0.83]	0.04 [0.02, 0.08]	0.15 [0.14, 0.17]
aerial duels	0.80 [0.74, 0.83]	0.06 [0.03, 0.12]	0.14 [0.13, 0.16]
dribbles	0.80 [0.70, 0.84]	0.11 [0.06, 0.21]	0.10 [0.09, 0.11]
received pass	0.78 [0.75, 0.80]	0.03 [0.02, 0.08]	0.18 [0.17, 0.20]
interceptions	0.77 [0.71, 0.80]	0.06 [0.03, 0.12]	0.17 [0.16, 0.19]
back passes	0.77 [0.75, 0.78]	0.01 [0.00, 0.02]	0.22 [0.21, 0.24]
successful passes	0.76 [0.73, 0.78]	0.03 [0.01, 0.06]	0.21 [0.19, 0.22]
passes to final third	0.76 [0.74, 0.78]	0.02 [0.01, 0.04]	0.22 [0.21, 0.24]
shot assists	0.74 [0.72, 0.76]	0.01 [0.01, 0.03]	0.24 [0.23, 0.26]
fouls suffered	0.73 [0.71, 0.75]	0.02 [0.01, 0.05]	0.25 [0.23, 0.26]
progressive run	0.72 [0.68, 0.75]	0.04 [0.02, 0.09]	0.23 [0.21, 0.25]
opponent half recoveries	0.69 [0.67, 0.71]	0.01 [0.00, 0.01]	0.31 [0.29, 0.33]
head shots	0.68 [0.66, 0.70]	0.00 [0.00, 0.00]	0.32 [0.30, 0.34]
defensive duels	0.68 [0.61, 0.71]	0.10 [0.05, 0.19]	0.23 [0.20, 0.25]
aerial duels won	0.66 [0.64, 0.68]	0.01 [0.00, 0.01]	0.33 [0.31, 0.35]
losses	0.64 [0.57, 0.68]	0.09 [0.05, 0.19]	0.26 [0.23, 0.28]
dribbles against	0.64 [0.56, 0.69]	0.14 [0.08, 0.25]	0.22 [0.19, 0.24]
goals	0.64 [0.62, 0.66]	0.00 [0.00, 0.00]	0.36 [0.34, 0.38]
key passes	0.63 [0.61, 0.65]	0.00 [0.00, 0.01]	0.36 [0.34, 0.38]
xg assist	0.63 [0.61, 0.65]	0.01 [0.00, 0.01]	0.36 [0.34, 0.38]
fouls	0.62 [0.59, 0.64]	0.03 [0.01, 0.07]	0.35 [0.32, 0.37]
loose ball duels	0.60 [0.51, 0.65]	0.15 [0.08, 0.28]	0.25 [0.21, 0.27]
accelerations	0.60 [0.51, 0.65]	0.14 [0.08, 0.26]	0.26 [0.22, 0.29]
shots blocked	0.59 [0.56, 0.61]	0.01 [0.01, 0.03]	0.40 [0.38, 0.42]
sliding tackles	0.56 [0.49, 0.61]	0.12 [0.06, 0.23]	0.32 [0.28, 0.35]
through passes	0.50 [0.42, 0.54]	0.14 [0.08, 0.27]	0.36 [0.31, 0.40]
own half losses	0.49 [0.46, 0.51]	0.02 [0.01, 0.06]	0.49 [0.46, 0.51]
offensive duels won	0.47 [0.45, 0.50]	0.02 [0.01, 0.05]	0.50 [0.48, 0.53]
pressing duels	0.44 [0.33, 0.52]	0.31 [0.20, 0.48]	0.25 [0.19, 0.29]
smart passes	0.40 [0.28, 0.49]	0.44 [0.30, 0.60]	0.17 [0.12, 0.21]

continued on next page

A. FULL INTRACLASSE-CORRELATION TABLE
(Table 1 continued from previous page)

Feature	ICC(player)	ICC(season)	ICC(resid.)
dangerous own half losses	0.39 [0.36, 0.42]	0.01 [0.01, 0.03]	0.60 [0.57, 0.62]
successful progressive passes	0.39 [0.34, 0.42]	0.09 [0.05, 0.19]	0.52 [0.46, 0.56]
yellow cards	0.36 [0.33, 0.39]	0.02 [0.01, 0.05]	0.61 [0.58, 0.65]
defensive duels won	0.36 [0.29, 0.40]	0.20 [0.11, 0.34]	0.44 [0.36, 0.50]
dangerous opponent half recoveries	0.35 [0.32, 0.38]	0.01 [0.00, 0.02]	0.64 [0.62, 0.67]
assists	0.33 [0.30, 0.35]	0.01 [0.00, 0.02]	0.66 [0.64, 0.69]
successful dribbles	0.30 [0.27, 0.33]	0.01 [0.01, 0.04]	0.69 [0.66, 0.72]

APPENDIX B

SQL extraction and construction queries

This appendix sketches the logic of the query used to extract and assemble the staged Wyscout data underlying the player–season dataset analysed in this master’s thesis. The real query runs to several hundred lines, dominated by long, mechanically repeated column lists (one line per raw statistic, one per per-90 feature, one per efficiency ratio); the sketch below reproduces its seven-stage logic without reproducing every column, so that the extraction and aggregation logic can be understood and, in principle, reproduced without reading several hundred near-identical lines. The complete, literal query is kept in the project repository at `appendix/sql/wyscout_extraction_query.sql` for anyone who needs it.

1. Extraction and construction logic

```
-- =====
-- Condensed sketch of the LaLiga player-season extraction and construction
-- query. Reproduces the logic of the seven stages below without the full
-- ~90-column select lists of the actual query, which is kept in this
-- directory (wyscout_extraction_query.sql) for anyone who needs the literal
-- SQL. Grain: one row per player_id x season_id x competition_id.
-- =====

with

-- 1) Competition and season scope -----
target_competitions as (
  select competition_id, competition_name, competition_area_name
  from stg_wyscout__competitions
  where lower(competition_name) like '%laliga%'
    and lower(coalesce(competition_gender, 'male')) = 'male'
    and coalesce(competition_division_level, 1) = 1 -- top flight only
),
target_seasons as (
  select season_id, season_name, competition_id
  from stg_wyscout__seasons
  where competition_id in (select competition_id from target_competitions)
),

-- 2) Match-player rows, restricted to the target competition/seasons -----
match_player_base as (
  select
    mps.player_id, mps.match_id, mps.season_id, mps.competition_id,
    p.player_name, p.role_name,
    mps.total_minutes_on_field,
    -- ~90 raw per-match counting statistics, grouped into blocks:
    -- attacking output (goals, shots, xG, touches in box, ...)
    -- passing and creation (passes, key passes, crosses, ...)
    -- carrying / 1v1 (dribbles, progressive runs, ...)
    -- defence and pressing (interceptions, recoveries, ...)
    -- duels (defensive/offensive/aerial duels, won/lost)
    -- discipline (fouls, cards, losses)
    -- plus a block of goalkeeper-specific markers kept only to drive the
    -- exclusion step below (total_gk_saves, total_gk_shots_against,
```

```

-- total_gk_clean_sheets, total_gk_exits, total_gk_conceded_goals)
mps.*
from stg_wyscout__matches_players_stats as mps
join target_seasons using (season_id)
join target_competitions using (competition_id)
left join stg_wyscout__players as p using (player_id)
),

-- 3) Goalkeeper exclusion, by behaviour rather than by role label -----
-- Role labels alone are not reliable (outfield players who briefly go in
-- goal, mislabelled positions); goalkeeper-specific activity is not.
outfield_match_player_base as (
  select *
  from match_player_base
  where coalesce(total_gk_shots_against, 0) = 0
    and coalesce(total_gk_saves, 0) = 0
    and coalesce(total_gk_conceded_goals, 0) = 0
    and coalesce(total_gk_clean_sheets, 0) = 0
    and coalesce(total_gk_exits, 0) = 0
),

-- 4) Aggregate match rows to one row per player x season -----
player_season_base as (
  select
    player_id, season_id, competition_id,
    max(player_name) as player_name,
    sum(total_minutes_on_field) as minutes,
    -- one sum() per row counting statistic carried over from step 2
    -- (goals, assists, passes, duels, cards, ... ~90 columns)
    sum(total_goals) as goals,
    sum(total_passes) as passes,
    sum(total_duels) as duels
    -- ... (remaining row counts omitted here, summed identically)
  from outfield_match_player_base
  group by player_id, season_id, competition_id
),

-- 5) Minimum-exposure filter -----
-- Drops small-sample player-seasons whose per-90 rates would otherwise be
-- dominated by noise.
player_season_filtered as (
  select *
  from player_season_base
  where minutes >= 600
),

-- 6) Feature construction: per-90 rates and efficiency ratios -----
player_season_features as (
  select
    psf.*,
    -- Per-90 intensity: one term per volume feature (~30 features total),
    -- covering attacking output, passing, carrying, defence, and duels.
    90.0 * psf.goals / nullif(psf.minutes, 0) as goals_p90,
    90.0 * psf.passes / nullif(psf.minutes, 0) as passes_p90,
    90.0 * psf.duels / nullif(psf.minutes, 0) as duels_p90,
    -- ... (remaining per-90 features constructed identically)

    -- Efficiency ratios: successful / attempted, one term per ratio
    -- feature (~18 features total: pass accuracy, dribble success rate,
    -- duel win rates by type, shot conversion, goal-per-shot, ...).
    1.0 * psf.successful_passes / nullif(psf.passes, 0) as pass_accuracy,
    1.0 * psf.duels_won / nullif(psf.duels, 0) as duel_win_rate
    -- ... (remaining ratio features constructed identically)

```

```
    from player_season_filtered as psf
)

-- 7) Final analytical output: one row per player-season -----
select * from player_season_features;
```

LISTING B.1. Condensed sketch of the LaLiga player-season extraction and construction query. Ellipsis comments mark places where the real query repeats the same pattern across additional columns.

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